Riemann Surfaces

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Complex analysis is full of functions that refuse to be functions. The logarithm is the obvious offender: exp is surjective onto $\mathbb{C}^*$ but wildly non-injective, so 'the' logarithm of z is only defined once we have made a choice, and no choice can be made consistently on all of $\mathbb{C}^*$. The same is true of $\sqrt{z}$, of $\sqrt{z^3 - z}$, and of the inverse of essentially any interesting holomorphic map. The traditional response is to slit the plane along a branch cut and work on what is left, but this is a confession of defeat: the cut is an artefact of our bookkeeping, not of the function, and the whole interest of log lies precisely in what happens when you walk around the origin and come back changed.

Riemann's idea was to stop mutilating the domain and start enlarging it. If log wants to take infinitely many values at each point of $\mathbb{C}^*$, then give it a domain with infinitely many points lying over each point of $\mathbb{C}^*$ – an infinite spiral ramp – on which it becomes an honest single-valued holomorphic function. The price is that this new domain is no longer an open subset of $\mathbb{C}$; it is a space which merely *looks like* $\mathbb{C}$ near each of its points. That is a Riemann surface, and once we allow ourselves such objects a great deal falls into place. Multi-valued functions become genuine functions, branch cuts become invisible, and – most strikingly – the analysis starts to control the topology. We will prove the Riemann–Hurwitz formula, which says that a holomorphic map between compact surfaces forces a numerical relation between their genera and its branching, and we will use it to compute the genus of a surface from nothing more than the equation that cuts it out.

This article constitutes my notes for the 'Riemann Surfaces' course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

Contents

§1 Holomorphic Functions Revisited

We begin by recalling, mostly without proof, what IB Complex Analysis gives us. Nothing in this section is new, but the results here are the local engine driving everything later on: every theorem we prove about Riemann surfaces will ultimately be a theorem about holomorphic functions on discs, transported by charts.

§1.1 Analytic Functions and their Zeros

Definition 1.1 (Domain)

A **domain** is a non-empty open connected subset $D \subseteq \mathbb{C}$.

Open discs $D(a, r) = \{z : |z - a| < r\}$, punctured discs $D(a, r) \setminus \{a\}$, annuli, half-planes and slit planes are all domains, and these are essentially the only ones we will ever need to write down explicitly.

Definition 1.2 (Holomorphic Function)

Let $D \subseteq \mathbb{C}$ be a domain. A function $f : D \to \mathbb{C}$ is **holomorphic** (or **analytic**) if it is complex differentiable at every point of D.

The great theorem of IB Complex Analysis is that this weak-looking hypothesis is enormously strong: a holomorphic function is automatically infinitely differentiable, and is represented by its Taylor series on any disc contained in its domain. In particular 'complex differentiable everywhere' and 'locally the sum of a convergent power series' mean the same thing, and we shall use the two descriptions interchangeably.[1]

The first consequence we need concerns zeros. A holomorphic function cannot vanish on a set with an accumulation point without vanishing identically – its zeros are isolated.

Proposition 1.3 (Isolated Zeros)

Let f be holomorphic on a domain D and suppose $f(z_0) = 0$. Then either f is identically zero on D, or there is a punctured disc about z_0 on which f does not vanish.

[1] This is why the words 'holomorphic' and 'analytic' are used as synonyms throughout. In the real setting they are very much not synonyms.

Proof. Write $f(z) = \sum_{n \geq 0} a_n (z - z_0)^n$ on some disc $U = D(z_0, r) \subseteq D$. If every a_n vanishes then $f \equiv 0$ on U, and a connectedness argument (the set where all derivatives of f vanish is open and closed in D) gives $f \equiv 0$ on all of D.

Otherwise let $m \geq 0$ be minimal with $a_m \neq 0$, so that on U we may write

$$f(z) = (z - z_0)^m g(z), \qquad g(z) = \sum_{n \geq 0} a_{m+n} (z - z_0)^n.$$

The series for g converges on U, so g is holomorphic there, and $g(z_0) = a_m \neq 0$. By continuity g is non-vanishing on some disc $D(z_0, \delta)$, and therefore f is non-vanishing on $D(z_0, \delta) \setminus \{z_0\}$. $\square$

The integer m appearing in the proof is the **order** of the zero of f at z_0, and it is worth noticing now – because the observation is the whole of Section 9 in disguise – that near such a point f looks like $z \mapsto z^m$ multiplied by a non-vanishing function.

Corollary 1.4 (Identity Theorem)
Let f, g be holomorphic on a domain D. If the set $\{z \in D : f(z) = g(z)\}$ has an accumulation point in D, then $f = g$ on D.

Proof. Apply Proposition 1.3 to $f - g$: the agreement set is not discrete, so $f - g$ cannot have isolated zeros, and hence vanishes identically. $\square$

This is the single most important fact in the course. It says a holomorphic function has no local freedom whatsoever: its germ at one point determines it everywhere on a connected domain. All of analytic continuation is an attempt to exploit this rigidity, and all of the trouble with analytic continuation comes from the word 'connected'.

§1.2 Singularities and Meromorphic Functions

Definition 1.5 (Isolated Singularity)
A function f holomorphic on a punctured disc $D(z_0, r) \setminus \{z_0\}$ is said to have an **isolated singularity** at z_0.

On such a punctured disc f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n,$$

converging locally uniformly, and the singularity is classified by the tail of negative coefficients.

Definition 1.6 (Types of Singularity)
With notation as above, we say z_0 is

(i) a **removable singularity** if $a_n = 0$ for all $n < 0$;

(ii) a **pole of order** $m > 0$ if $a_{-m} \neq 0$ and $a_n = 0$ for all $n < -m$;

(iii) an **essential singularity** otherwise.

A pole of order 1 is called **simple**.

Theorem 1.7 (Riemann Removable Singularity Theorem)
If f is holomorphic and bounded on a punctured disc about z_0, then z_0 is a removable singularity, and f extends holomorphically over z_0.

Theorem 1.8 (Casorati–Weierstrass)
The point z_0 is an essential singularity of f if and only if $f(U)$ is dense in $\mathbb{C}$ for every punctured neighbourhood U of z_0.

Both were proved in IB Complex Analysis and we take them as given. The trichotomy is worth restating in the form we shall actually use: as $z \to z_0$, either $f(z)$ tends to a limit in $\mathbb{C}$ (removable), or $|f(z)| \to \infty$ (pole), or f oscillates so violently that it comes arbitrarily close to every complex number (essential). Poles are therefore not really singularities at all if we are willing to let ∞ count as a value, and that willingness is exactly what the Riemann sphere will formalise.

Definition 1.9 (Meromorphic Function)
Let D be a domain and $A \subseteq D$ a discrete closed subset. A holomorphic function $f : D \setminus A \to \mathbb{C}$ with a pole at each point of A is called **meromorphic** on D.

Example 1.10
The function

$$f(z) = \frac{1}{e^{1/z} - 1}$$

is meromorphic on the upper half-plane, with simple poles at the points $1/(2\pi i n)$ for $n \in \mathbb{N}$. Note what happens if we try to regard f as meromorphic on all of $\mathbb{C}$: the poles accumulate at 0, so the set of poles is no longer discrete as a subset of $\mathbb{C}$, and indeed 0 is an essential singularity of f. The condition that A be discrete *and closed in* D is doing real work.

§1.3 The Open Mapping Theorem and the Maximum Modulus Principle

The last two facts we shall want are global in flavour.

Theorem 1.11 (Open Mapping Theorem)
A non-constant holomorphic function on a domain $D \subseteq \mathbb{C}$ is an open map.

Theorem 1.12 (Maximum Modulus Principle)
Let f be holomorphic on a domain D. If $|f|$ attains a local maximum at some point of D, then f is constant.

The second follows immediately from the first: if $f(D)$ is open then no point of $f(D)$

can be a point of maximum modulus of $f(D)$, since every neighbourhood of a point w contains points of larger modulus.

Finally we record the local statement that will become the definition of multiplicity.

Theorem 1.13 (Inverse Function Theorem)

Let f be holomorphic near z_0 with $f'(z_0) \neq 0$. Then there are open neighbourhoods $U \ni z_0$ and $V \ni f(z_0)$ such that $f|_U : U \to V$ is a biholomorphism, i.e. a holomorphic bijection with holomorphic inverse.

§2 Analytic Continuation

We now come to the problem that motivates the whole course. The identity theorem says that a holomorphic function on a domain is determined by its restriction to any small disc. Turning this around: given a holomorphic function on a small disc, can we recover 'the' function it came from, on some larger domain? And if we can do it in more than one way, how do the answers differ?

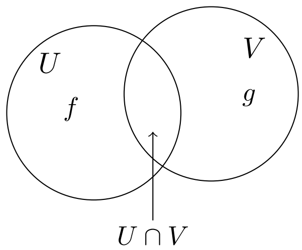
§2.1 Function Elements and Direct Continuation

Definition 2.1 (Function Element)

Let $D \subseteq \mathbb{C}$ be a domain. A **function element** on D is a pair (f, U) where $U \subseteq D$ is a subdomain and $f : U \to \mathbb{C}$ is holomorphic.

Definition 2.2 (Direct Analytic Continuation)

Two function elements (f, U) and (g, V) on D are **direct analytic continuations** of one another, written $(f, U) \sim (g, V)$, if $U \cap V \neq \emptyset$ and $f = g$ on $U \cap V$.



By the identity theorem, if $U \cap V$ is connected then g is completely determined by f (and by the choice of V): there is at most one holomorphic g on V agreeing with f on $U \cap V$. If $U \cap V$ is disconnected the definition still makes sense, but we should be careful, since agreeing on one component is then weaker than agreeing on all of them. We will normally arrange for our sets to be discs, so that intersections are convex and this issue evaporates.

The relation $\sim$ is reflexive and symmetric but emphatically *not* transitive – that failure is the entire subject. So we take its transitive closure.

Definition 2.3 (Analytic Continuation)

We write $(f, U) \approx (g, V)$ if there is a finite chain of function elements

$$(f, U) = (f_0, U_0) \sim (f_1, U_1) \sim \cdots \sim (f_n, U_n) = (g, V)$$

on D. We then say (g, V) is an **analytic continuation** of (f, U).

Now $\approx$ *is* an equivalence relation, being the transitive closure of a reflexive symmetric relation.

Definition 2.4 (Complete Analytic Function)

An equivalence class $\mathcal{F}$ of function elements on D under $\approx$ is called a **complete analytic function** on D.

The name is a little grandiose for what is, at this stage, just a set of pairs. But the point of view is the right one: a 'multi-valued function' is not a function with several values, it is a *collection of ordinary functions, glued along their overlaps.* Our job over the next few sections is to make the gluing literal.

§2.2 The Complex Logarithm

Let us do the fundamental example completely explicitly. Write $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$, and recall that $\exp : \mathbb{C} \to \mathbb{C}^*$ is surjective with $\exp(z) = \exp(w)$ if and only if $z - w \in 2\pi i\mathbb{Z}$. We want to invert it.

For an interval $(\alpha, \beta) \subseteq \mathbb{R}$ with $\beta - \alpha \leq 2\pi$, set

$$U_{(\alpha,\beta)} = \{re^{i\theta} : r > 0,\ \alpha < \theta < \beta\},$$

an open sector, and define $f_{(\alpha,\beta)} : U_{(\alpha,\beta)} \to \mathbb{C}$ by

$$f_{(\alpha,\beta)}(re^{i\theta}) = \log r + i\theta, \qquad \theta \in (\alpha, \beta).$$

This is well defined precisely because the angle θ of a point of $U_{(\alpha,\beta)}$ is unique in the range (α, β), and it is holomorphic since it is a local inverse of exp, which has non-vanishing derivative. So

$$F_{(\alpha,\beta)} = (f_{(\alpha,\beta)}, U_{(\alpha,\beta)})$$

is a function element on $\mathbb{C}^*$, and $\exp \circ f_{(\alpha,\beta)} = \mathrm{id}$.

To keep the bookkeeping finite, restrict attention to quarter-turn sectors: for $n \in \mathbb{Z}$ put

$$I(n) = \left(\frac{(n-1)\pi}{2}, \frac{(n+1)\pi}{2}\right),$$

an interval of length π centred on $n\pi/2$, and abbreviate $U_n = U_{I(n)}$, $f_n = f_{I(n)}$, $F_n = F_{I(n)}$. Thus U_0 is the right half-plane, U_1 the upper half-plane, and so on; U_n depends only on n modulo 4, but f_n does not.

Proposition 2.5

$F_m \sim F_n$ if and only if $|m - n| \leq 1$.

Proof. First suppose $|m-n| \leq 1$. If $m = n$ there is nothing to do, so take $n = m+1$. Then $U_m \cap U_{m+1}$ is the open sector of angles in $(m\pi/2, (m+1)\pi/2)$, which is non-empty, and for $z = re^{i\theta}$ in it we may take θ in that common range, whence $f_m(z) = \log r + i\theta = f_{m+1}(z)$. So $F_m \sim F_{m+1}$.

Conversely suppose $|m-n| \geq 2$. If $|m-n| \geq 4$ then $I(m)$ and $I(n)$ are disjoint modulo nothing at all – more precisely the sectors U_m, U_n may still meet, but any z in the intersection has $f_m(z) - f_n(z) \in 2\pi i\mathbb{Z} \setminus \{0\}$, since the two determinations of the argument differ by a non-zero multiple of 2π. If $|m-n| = 2$ then U_m and U_n are opposite half-planes meeting in two opposite quadrants, and $|m-n| = 3$ gives an intersection which is a single quadrant; in every case a point of the intersection has $f_m(z) \neq f_n(z)$, because the arguments used differ by a non-zero multiple of 2π. Hence $F_m \not\sim F_n$. □

Corollary 2.6

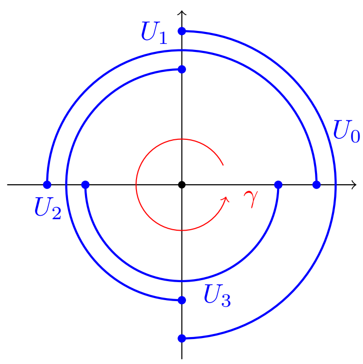
$F_m \approx F_n$ for all $m, n \in \mathbb{Z}$.

Proof. Chain along $F_m \sim F_{m\pm1} \sim \cdots \sim F_n$. □

So all the F_n lie in a single complete analytic function on $\mathbb{C}^*$, which we may reasonably call *the* complex logarithm. Notice what has happened: $U_0 = U_4$ as subsets of $\mathbb{C}$, but

$$f_0(1) = 0, \qquad f_4(1) = 2\pi i.$$

Continuing once round the origin has changed the function. Analytic continuation is therefore *not* path-independent, and the complete analytic function log is genuinely bigger than any one of its elements.



Each U_n is a half-plane, drawn here as a semicircular arc; the four arcs are given different radii only so that they can be told apart, and the sector boundaries are the two coordinate axes. Together the $U_0, \ldots, U_3$ cover $\mathbb{C}^*$, and each overlaps the next in a quadrant. Continuing f_0 around the loop γ takes us through F_1, F_2, F_3 and back to U_0, but with $f_4 = f_0 + 2\pi i$.

§2.3 Complex Roots

Exactly the same analysis handles k-th roots. Let $p_k : \mathbb{C}^* \to \mathbb{C}^*$ be $z \mapsto z^k$ for $k \geq 2$, and define, on the same sectors U_n as above,

$$g_n(z) = \exp\left(\tfrac{1}{k} f_n(z)\right), \qquad G_n = (g_n, U_n).$$

Then $g_n(z)^k = \exp(f_n(z)) = z$, so each g_n is a holomorphic branch of the k-th root on U_n.

Since $f_{n+4} = f_n + 2\pi i$ we get $g_{n+4} = e^{2\pi i/k} g_n$, and more generally g_n depends only on n modulo $4k$. Repeating the case analysis of Proposition 2.5 shows that $G_m \sim G_n$ if and only if $m \equiv n$ or $m \equiv n \pm 1$ modulo $4k$, and hence all the G_n lie in a single complete analytic function. This time, though, going round the origin *four* times returns us to $e^{2\pi i/k} g_0$, and it takes k circuits to get back to where we started. The multi-valuedness of $\sqrt[k]{\cdot}$ is finite, of order k, where that of log was infinite.

§2.4 Natural Boundaries

Before building surfaces, we pause on an obstruction: sometimes a function element admits no continuation at all past the boundary of its natural domain.

Write $\mathbb{D} = D(0,1)$ and $\mathbb{T} = \partial\mathbb{D}$. Let $f(z) = \sum_{n\geq 0} a_n z^n$ be a power series with radius of convergence exactly 1, so that $(f, \mathbb{D})$ is a function element.

Definition 2.7 (Regular and Singular Points)

A point $z_0 \in \mathbb{T}$ is **regular** for f if there is an open neighbourhood U of z_0 and a function element (g, U) with $g = f$ on $U \cap \mathbb{D}$. Otherwise z_0 is **singular** for f.

The set of regular points is visibly open in $\mathbb{T}$, so the singular points form a closed set. It is tempting to think that z_0 is regular exactly when the power series converges at z_0, and this is false in both directions.

Example 2.8

Take $f(z) = 1/(1-z) = \sum_{n\geq 0} z^n$. The only singular point is $z = 1$, so $z = -1$ is regular; but the power series $\sum(-1)^n$ does not converge at -1. Convergence is not necessary for regularity.

Example 2.9

Take $g(z) = \sum_{n\geq 2} \frac{z^n}{n(n-1)}$. This converges absolutely at $z = 1$. But $g''(z) = \sum_{n\geq 2} z^{n-2} = 1/(1-z)$, which is singular at 1; so g cannot be continued past 1 either, and 1 is singular for g. Convergence is not sufficient for regularity.

Proposition 2.10

If $f(z) = \sum_{n\geq 0} a_n z^n$ has radius of convergence exactly 1, then some point of $\mathbb{T}$ is singular for f.

Proof. Suppose not. Then every $z_0 \in \mathbb{T}$ has a disc $D(z_0, \varepsilon_{z_0})$ carrying a continuation of f. By compactness finitely many such discs cover $\mathbb{T}$; by the identity theorem (the intersections being convex, hence connected) these continuations agree with each other and with f on overlaps, and so patch together with f to give a holomorphic function on an open set containing $\overline{\mathbb{D}}$. That set contains $D(0, 1+\delta)$ for some $\delta > 0$, again by compactness, and so the Taylor series of f at 0 would have radius of convergence at least $1 + \delta$, a contradiction. $\square$

Definition 2.11 (Natural Boundary)

If every point of $\mathbb{T}$ is singular for f, we call $\mathbb{T}$ the **natural boundary** of f.

Example 2.12 (A Lacunary Series)

Let $f(z) = \sum_{n \geq 0} z^{n!}$, which has radius of convergence 1. We claim $\mathbb{T}$ is its natural boundary.

Since the singular set is closed, it suffices to show that the roots of unity are all singular, as these are dense in $\mathbb{T}$. So let $\omega = e^{2\pi i p/q}$ with $p, q \in \mathbb{Z}$, $q \geq 1$. For $n \geq q$ we have $q \mid n!$ and so $\omega^{n!} = 1$. Hence for $r \in (0, 1)$,

$$f(r\omega) = \sum_{n=0}^{q-1} r^{n!}\omega^{n!} + \sum_{n \geq q} r^{n!}.$$

The first sum is a finite sum of terms of modulus at most 1, hence bounded by q. For the second, given any $M > 0$ we have

$$\sum_{n \geq q} r^{n!} \geq \sum_{n=q}^{q+M} r^{n!} \longrightarrow M + 1$$

as $r \to 1^-$, so $\sum_{n \geq q} r^{n!} > M$ for r close enough to 1. Therefore $f(r\omega) \to \infty$ as $r \to 1^-$. A function element (g, U) agreeing with f near ω would be bounded near ω, so no such element exists and ω is singular.

So continuation can fail entirely. When it does not fail, we would like to understand how the answer depends on the route taken – which requires us to say what a route is.

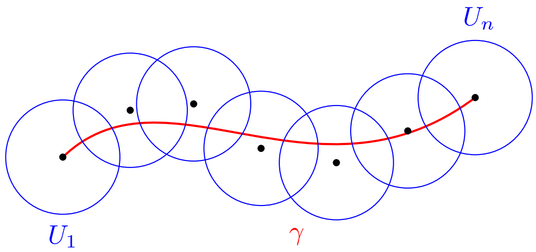
§2.5 Continuation Along a Path

Definition 2.13 (Continuation Along a Path)

Let $\gamma : [0, 1] \to D$ be a path and let $(f, U), (g, V)$ be function elements on D with $\gamma(0) \in U$ and $\gamma(1) \in V$. We write $(f, U) \approx_\gamma (g, V)$ if there is a dissection $0 = t_0 < t_1 < \cdots < t_n = 1$ and function elements

$$(f, U) = (f_1, U_1),\ (f_2, U_2),\ \ldots,\ (f_n, U_n) = (g, V)$$

with $(f_i, U_i) \sim (f_{i+1}, U_{i+1})$ for each i, and $\gamma([t_{i-1}, t_i]) \subseteq U_i$ for each i. We then say (f, U) can be **continued along** γ, with result (g, V).



The picture is a chain of discs strung along γ, each carrying a holomorphic function agreeing with its neighbours. Definition 2.13 is a little unwieldy, and we will replace it in Section 7 by something much cleaner: continuation along γ will simply be *lifting γ to the space of germs*. Once we have said that, the key uniqueness statement – the monodromy theorem, that the result of continuing along γ depends only on the homotopy class of γ – becomes a special case of the homotopy lifting property from IB/II Algebraic Topology.

§3 The Surfaces of $\log$ and $\sqrt[k]{z}$

Before setting up the general theory, let us build the two surfaces the theory is meant to explain. This is worth doing by hand: everything we do abstractly later is modelled on these two constructions.

§3.1 Gluing the Sectors

Recall from Section 2.2 the function elements $F_n = (f_n, U_n)$ for log. The trouble was that U_0 and U_4 are the same subset of $\mathbb{C}$ but carry different functions. So let us refuse to identify them. Define

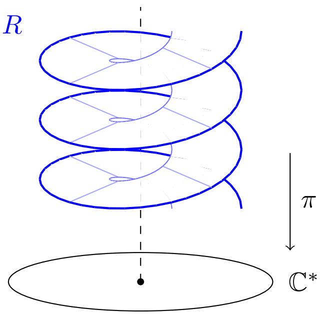
$$R = \left(\coprod_{n \in \mathbb{Z}} U_n \right) \Big/ \sim,$$

where a point z_1 in the copy U_m is identified with a point z_2 in the copy U_n if and only if

$$z_1 = z_2 \text{ as elements of } \mathbb{C} \quad \text{and} \quad f_m(z_1) = f_n(z_2),$$

and give R the quotient topology. In words: we lay out infinitely many copies of the sectors, and glue copy m to copy n exactly where the two branches of log agree. By Proposition 2.5 this happens exactly when $|m - n| \leq 1$, so each sector is glued to its two neighbours along a quadrant, and the result spirals.

The picture to have in mind is an infinite multi-storey car park: a helical ramp winding endlessly up and down over the punctured plane, with no floor ever meeting another except its immediate neighbours.



Two maps come along for free. First, the branches assemble into a single global function: define $f : R \to \mathbb{C}$ by $f([z]) = f_n(z)$ for $z \in U_n$. This is well defined precisely because we built the equivalence relation to make it so. Second, the inclusions $U_n \hookrightarrow \mathbb{C}^*$ assemble into $\pi : R \to \mathbb{C}^*$, $\pi([z]) = z$, which is well defined because equivalent points are equal in $\mathbb{C}$. These satisfy

$$\exp \circ f = \pi,$$

which says exactly that f deserves the name 'logarithm': applying exp to f recovers the point of $\mathbb{C}^*$ underneath.

Proposition 3.1

R is path-connected and Hausdorff.

Proof. Path-connectedness: each U_n is connected, and U_n meets U_{n+1} in R (as $F_n \sim F_{n+1}$), so the images of the U_n in R form a chain of overlapping connected sets covering R.

For Hausdorffness, consider $\Phi : R \to \mathbb{C}^2$ given by

$$\Phi([z]) = (\pi([z]), f([z])).$$

This is continuous, and it is injective: if $\Phi([z_1]) = \Phi([z_2])$ then $z_1 = z_2$ in $\mathbb{C}$ and the two branch values agree, which is precisely the condition defining $\sim$. A continuous injection into a Hausdorff space separates points, so R is Hausdorff. □

Remark. In fact Φ identifies R with the graph

$$\{(w, z) \in \mathbb{C}^2 : w = \exp z\},$$

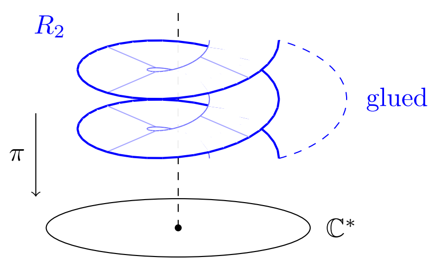
and under this identification π and f are the two coordinate projections. So the Riemann surface of log is nothing more exotic than the graph of exp with the roles of the axes swapped. This is a useful sanity check, and a preview: in general, the Riemann surface of an algebraic function will be (a compactification of) the graph of the equation defining it.

§3.2 The Surface of the k-th Root

The same construction with the elements $G_n = (g_n, U_n)$ produces a space R_k. Since g_n depends only on n modulo $4k$, this time only $4k$ copies of the sectors are involved and the spiral closes up after k turns. We again get $g : R_k \to \mathbb{C}^*$ and $\pi : R_k \to \mathbb{C}^*$ with

$$p_k \circ g = \pi, \qquad p_k(z) = z^k,$$

and, exactly as before, R_k is path-connected and Hausdorff. Here is the picture for $k = 2$: a two-sheeted spiral, closing after two turns.



In both cases g (respectively f) is the honest single-valued function we were after, and π is the map remembering which point of $\mathbb{C}^*$ we are sitting over. The key structural feature of π is that it is a *local homeomorphism*: each little piece of R maps homeomorphically to a piece of $\mathbb{C}^*$. That is what will let us transport the complex structure of $\mathbb{C}^*$ upstairs, and it is where we turn next.

Remark. One thing is unsatisfactory about R_k: the interesting point 0 is missing, and so is ∞. We would like to add them, obtaining a map $\hat{p}_k : \mathbb{C}_\infty \to \mathbb{C}_\infty$ which is k-to-1 everywhere except at the two points $0, \infty$, where it is 1-to-1. These exceptional points are the **branch points**, and understanding them is the goal of Section 9. Once the points are added, π is no longer a covering map – but it is still holomorphic, and that turns out to be the more useful category.

§4 Riemann Surfaces

§4.1 Covering Maps

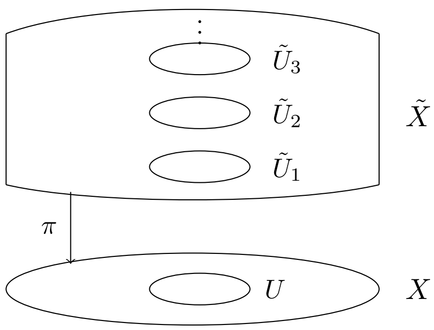
We isolate the topological property of $\pi : R \to \mathbb{C}^*$ that made the last section work. This material overlaps heavily with II Algebraic Topology; we repeat it because the conventions differ slightly and because we shall need to be careful about exactly which hypotheses are used where.

Definition 4.1 (Covering Map)

Let $\tilde{X}, X$ be path-connected Hausdorff spaces. A continuous surjection $\pi : \tilde{X} \to X$ is a **covering map** if it is a local homeomorphism: every $\tilde{x} \in \tilde{X}$ has an open neighbourhood $\tilde{U}$ such that $\pi|_{\tilde{U}}$ is a homeomorphism onto an open subset of X.

Definition 4.2 (Regular Covering Map)

A covering map $\pi : \tilde{X} \to X$ is **regular** if every $x \in X$ has an open neighbourhood U which is **evenly covered**: there is a discrete set Δ and a homeomorphism $\pi^{-1}(U) \cong U \times \Delta$ under which π becomes the projection $(u, \delta) \mapsto u$. Equivalently, $\pi^{-1}(U)$ is a disjoint union of open sets, the **sheets** over U, each mapped homeomorphically onto U by π.



Regularity is a genuinely stronger condition, and it is the one that matters for lifting paths. The distinction is easy to miss, so here are the examples we shall use.

Example 4.3

(i) The map $\pi : R \to \mathbb{C}^*$ of Section 3 is a regular covering map. Indeed

$$\pi^{-1}(U_n) = \coprod_{m \equiv n \ (4)} U_m \cong U_n \times \mathbb{Z},$$

since the copy U_m maps homeomorphically to U_n exactly when $m \equiv n$ modulo 4, and distinct such copies are disjoint in R.

(ii) $\exp : \mathbb{C} \to \mathbb{C}^*$ is a regular covering map. For an open interval $I \subseteq \mathbb{R}$ of length at most 2π, write $\tilde{V}_I = \mathbb{R} + iI$, a horizontal strip; then exp restricts to a homeomorphism $\tilde{V}_I \to U_I$ with inverse a branch of log, and

$$\exp^{-1}(U_{I(n)}) = \coprod_{m \equiv n \ (4)} \tilde{V}_{I(m)} \cong U_{I(n)} \times \mathbb{Z}.$$

(iii) $p_k : \mathbb{C}^* \to \mathbb{C}^*$, $z \mapsto z^k$, is a regular covering map with k sheets: over the sector U_I with I of length less than $2\pi/k$ sit the k sectors $\zeta^j U_{I/k}$, $\zeta = e^{2\pi i/k}$.

(iv) *(Non-example.)* The inclusion $\iota : \mathbb{D} \hookrightarrow \mathbb{C}$ is a local homeomorphism, hence a covering map onto its image in the weak sense above, but it is not regular as a map to $\mathbb{C}$: for $z_0 \in \mathbb{T}$ and any neighbourhood U of z_0, we have $\iota^{-1}(U) = U \cap \mathbb{D}$, whose image is not U. More basically, ι is not surjective. Regularity is what rules out this kind of 'partial' cover.

§4.2 Charts and Atlases

Now we say what a surface is. The idea is Riemann's: a space that looks locally like an open subset of $\mathbb{C}$, in a way that is compatible from patch to patch.

Definition 4.4 (Chart)

Let R be a topological space. A **chart** on R is a pair (ϕ, U) where $U \subseteq R$ is open and $\phi : U \to \phi(U) \subseteq \mathbb{C}$ is a homeomorphism onto an open subset of $\mathbb{C}$. We call ϕ a **local coordinate** on U.

Definition 4.5 (Atlas)

A collection $\mathcal{A}$ of charts on R is an **atlas** if

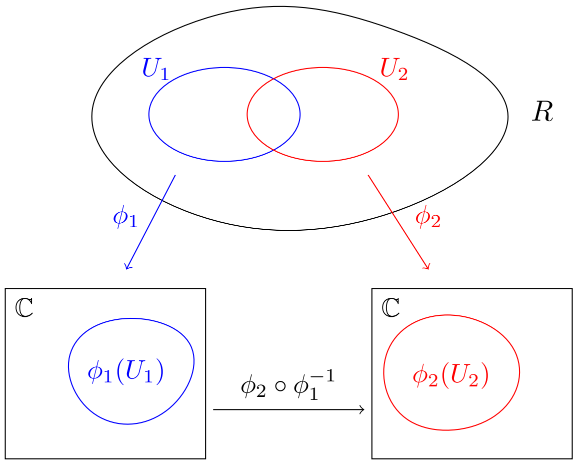
(i) $\bigcup_{(\phi,U)\in\mathcal{A}} U = R$;

(ii) for all $(\phi_1, U_1), (\phi_2, U_2) \in \mathcal{A}$ with $U_1 \cap U_2 \neq \emptyset$, the **transition function**

$$\phi_1 \circ \phi_2^{-1} : \phi_2(U_1 \cap U_2) \to \phi_1(U_1 \cap U_2)$$

is holomorphic.

Note that the transition function goes between open subsets of $\mathbb{C}$, so 'holomorphic' means what it always meant. Note also that its inverse is $\phi_2 \circ \phi_1^{-1}$, which is holomorphic by the same condition with the roles swapped; so transition functions are automatically biholomorphic.



There is an annoying redundancy here, which the next example exposes.

Example 4.6

Take $R = \mathbb{C}$. Let $\mathcal{A}$ consist of all pairs (id_U, U) for $U \subseteq \mathbb{C}$ open; this is an atlas. Let $\mathcal{A}'$ consist of all pairs $(z \mapsto z+1, U)$; this too is an atlas. So is $\mathcal{A} \cup \mathcal{A}'$, since the transition functions are translations. Yet all three describe the same complex analysis on $\mathbb{C}$.

The cure is to take the largest possible atlas, so that no information is hidden in a choice.

Definition 4.7 (Conformal Structure)

A **conformal structure** on R is an atlas $\mathcal{A}$ which is **maximal**: whenever a chart (ψ, V) on R has the property that $\phi \circ \psi^{-1}$ is holomorphic (where defined) for every $(\phi, U) \in \mathcal{A}$, we have $(\psi, V) \in \mathcal{A}$.

Definition 4.8 (Riemann Surface)

A **Riemann surface** is a pair $(R, \mathcal{A})$ where R is a path-connected Hausdorff topological space and $\mathcal{A}$ is a conformal structure on R.

We will normally suppress $\mathcal{A}$ and just write R. Note that connectedness is part of the definition here – some authors allow disconnected Riemann surfaces, but for us a Riemann surface is connected, which makes the identity theorem available at all times.

Maximal atlases are unwieldy to write down, so in practice we always specify a small atlas and appeal to the following.

Lemma 4.9

Every atlas $\mathcal{A}$ on R is contained in a unique conformal structure $\hat{\mathcal{A}}$.

Proof. There is only one possible candidate: let $\hat{\mathcal{A}}$ be the set of all charts (ψ, V)

on R such that $\psi \circ \phi^{-1}$ and $\phi \circ \psi^{-1}$ are holomorphic for every $(\phi, U) \in \mathcal{A}$. Any conformal structure containing $\mathcal{A}$ is contained in $\hat{\mathcal{A}}$ by definition of an atlas, and contains $\hat{\mathcal{A}}$ by maximality; so uniqueness is clear once we know $\hat{\mathcal{A}}$ is an atlas.

It covers R since $\mathcal{A} \subseteq \hat{\mathcal{A}}$ does. For compatibility, take $(\psi_1, V_1), (\psi_2, V_2) \in \hat{\mathcal{A}}$ and $p \in V_1 \cap V_2$. Choose $(\phi, U) \in \mathcal{A}$ with $p \in U$. Near $\psi_2(p)$ we may write

$$\psi_1 \circ \psi_2^{-1} = (\psi_1 \circ \phi^{-1}) \circ (\phi \circ \psi_2^{-1}),$$

a composite of holomorphic maps, hence holomorphic at $\psi_2(p)$. As p was arbitrary, $\psi_1 \circ \psi_2^{-1}$ is holomorphic on $\psi_2(V_1 \cap V_2)$. Maximality of $\hat{\mathcal{A}}$ is immediate from its definition. □

So from now on, 'here is an atlas' means 'here is a Riemann surface'.

§4.3 First Examples

Example 4.10 (The Plane)

Taking $\mathcal{A} = \{(\mathrm{id}, \mathbb{C})\}$ and extending gives the **canonical conformal structure** on $\mathbb{C}$. Whenever we speak of $\mathbb{C}$ as a Riemann surface, this is what is meant. Note that by Example 4.6 the atlas of all translations extends to the same structure.

Example 4.11 (The Conjugate Plane)

The single chart $(z \mapsto \bar{z}, \mathbb{C})$ is also an atlas on $\mathbb{C}$, and it does *not* extend to the canonical structure, since $z \mapsto \bar{z}$ is not holomorphic. Write $\bar{\mathbb{C}}$ for $\mathbb{C}$ with this structure. So the same topological space can carry different conformal structures. (As it happens $\bar{\mathbb{C}} \cong \mathbb{C}$ as Riemann surfaces, via $z \mapsto \bar{z}$ – but they are not *equal*.)

Example 4.12 (Open Subsets)

If R is a Riemann surface with conformal structure $\mathcal{A}$ and $S \subseteq R$ is open and connected, then

$$\{(\phi|_{U \cap S}, U \cap S) : (\phi, U) \in \mathcal{A}\}$$

is an atlas on S, making S a Riemann surface. In particular every domain $D \subseteq \mathbb{C}$ – the disc $\mathbb{D}$, the upper half-plane $\mathbb{H}$, the punctured plane $\mathbb{C}^*$ – is a Riemann surface.

Example 4.13 (The Riemann Sphere)

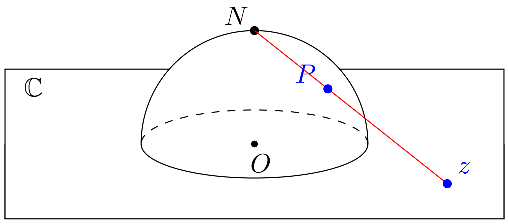
Let $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$ with the topology in which neighbourhoods of ∞ are the complements of compact subsets of $\mathbb{C}$. Take the two charts

$$(\phi_0, U_0) = (\mathrm{id}, \mathbb{C}), \qquad (\phi_1, U_1) = (z \mapsto 1/z,\ \mathbb{C}^* \cup \{\infty\}),$$

where we read $1/\infty = 0$. Each is a homeomorphism onto $\mathbb{C}$, they cover $\mathbb{C}_\infty$, and on the overlap $U_0 \cap U_1 = \mathbb{C}^*$ the transition function is $z \mapsto 1/z$, which is holomorphic there. So this is an atlas, and $\mathbb{C}_\infty$ is a Riemann surface, the **Riemann sphere**.

That $\mathbb{C}_\infty$ deserves to be called a sphere is the content of stereographic projection. Identify $\mathbb{C}$ with the plane $\{x_3 = 0\}$ in $\mathbb{R}^3$ and let S^2 be the unit sphere, with north pole

$N = (0, 0, 1)$. Each point $P \in S^2 \setminus \{N\}$ determines a unique point of the plane, namely where the line NP meets it, and this gives a homeomorphism $S^2 \setminus \{N\} \to \mathbb{C}$; sending $N \mapsto \infty$ extends it to a homeomorphism $S^2 \to \mathbb{C}_\infty$.



Chasing the geometry, if $P = (x_1, x_2, x_3)$ then the line $N + t(P - N)$ meets $x_3 = 0$ at $t = 1/(1 - x_3)$, giving

$$z = \frac{x_1 + ix_2}{1 - x_3}.$$

Projecting instead from the south pole $S = (0, 0, -1)$ gives $z' = (x_1 - ix_2)/(1 + x_3)$, and one checks $zz' = 1$ using $x_1^2 + x_2^2 + x_3^2 = 1$. So the two stereographic projections differ by $z \mapsto 1/z$ – exactly the transition function of Example 4.13. The atlas we wrote down is the geometrically natural one.

Remark. Under this identification, '$z \to \infty$' means '$P \to N$', so the point ∞ is in no way special: the sphere has a transitive group of conformal automorphisms (the Möbius transformations) and ∞ can be moved anywhere. That is the real payoff of the Riemann sphere as against the extended plane of a first course.

§4.4 Holomorphic Maps

Definition 4.14 (Holomorphic Map)

Let R, S be Riemann surfaces. A continuous map $f : R \to S$ is **holomorphic** (or **analytic**) if for every chart (ϕ, U) on R and every chart (ψ, V) on S, the map

$$\psi \circ f \circ \phi^{-1}$$

is holomorphic on $\phi(U \cap f^{-1}(V))$.

The composite $\psi \circ f \circ \phi^{-1}$ is called the **local expression** of f in the charts ϕ and ψ. Checking the condition for *all* charts is of course impossible in practice; happily it suffices to check one chart at each point.

Lemma 4.15

A continuous map $f : R \to S$ is holomorphic if and only if for each $p \in R$ there exist charts (ϕ_p, U_p) on R with $p \in U_p$ and (ψ_p, V_p) on S with $f(p) \in V_p$ such that $\psi_p \circ f \circ \phi_p^{-1}$ is holomorphic on $\phi_p(U_p \cap f^{-1}(V_p))$.

Proof. Necessity is trivial. For sufficiency, let (ϕ, U) and (ψ, V) be arbitrary charts; we must show $\psi \circ f \circ \phi^{-1}$ is holomorphic at $\phi(p)$ for each $p \in U \cap f^{-1}(V)$. Take $(\phi_p, U_p), (\psi_p, V_p)$ as hypothesised. On a small enough neighbourhood of $\phi(p)$ we may factor

$$\psi \circ f \circ \phi^{-1} = (\psi \circ \psi_p^{-1}) \circ (\psi_p \circ f \circ \phi_p^{-1}) \circ (\phi_p \circ \phi^{-1}).$$

The outer two factors are transition functions, hence holomorphic; the middle factor is holomorphic by hypothesis. So the composite is holomorphic at $\phi(p)$. □

Lemma 4.16

If $f : R \to S$ and $g : S \to T$ are holomorphic, so is $g \circ f : R \to T$.

Proof. Immediate from Lemma 4.15: locally $g \circ f$ has local expression $(\chi \circ g \circ \psi^{-1}) \circ (\psi \circ f \circ \phi^{-1})$. □

Definition 4.17 (Conformal Equivalence)

A **conformal equivalence** (or **biholomorphism**) is a holomorphic bijection $f : R \to S$ whose inverse is holomorphic. We then write $R \cong S$.

By Lemma 4.16 this is an equivalence relation on Riemann surfaces, and it is the notion of sameness we care about. We will see later (Corollary 9.12) that the hypothesis on f^{-1} is automatic: a holomorphic bijection of Riemann surfaces is always a conformal equivalence.

Example 4.18

The map $\mathbb{C} \to \bar{\mathbb{C}}$, $z \mapsto \bar{z}$, is a conformal equivalence: in the chart id upstairs and $z \mapsto \bar{z}$ downstairs, its local expression is $z \mapsto \bar{\bar{z}} = z$.

Example 4.19 (Möbius Maps)

For $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$, the Möbius transformation

$$\mu(z) = \frac{az + b}{cz + d}, \qquad \mu(-d/c) = \infty, \quad \mu(\infty) = a/c$$

is a conformal equivalence $\mathbb{C}_\infty \to \mathbb{C}_\infty$. Indeed μ is holomorphic away from $-d/c$ and ∞, and at those points one checks holomorphy in the chart $1/z$: for example near $z = \infty$ the local expression is $w \mapsto (cw + d)/(aw + b)$ (in the chart $w = 1/z$ upstairs and $1/z$ downstairs when $a \neq 0$), which is holomorphic at 0. The inverse is again Möbius. In fact *every* conformal automorphism of $\mathbb{C}_\infty$ is of this form, as we shall see.

Example 4.20 ($\mathbb{C}^*$ and the Cylinder)

The map $\mathbb{C} \to \mathbb{C}^*$, $z \mapsto e^z$, is holomorphic and surjective but not injective; it descends to a conformal equivalence $\mathbb{C}/2\pi i\mathbb{Z} \to \mathbb{C}^*$ once we know how to give the quotient a conformal structure. That is the next subsection.

§4.5 Structures Induced by Covering Maps

Here is the tool that lets us construct nearly all our examples: a covering of a Riemann surface is a Riemann surface, in exactly one way.

Lemma 4.21

Let R be a Riemann surface and $\pi : \tilde{R} \to R$ a covering map, with $\tilde{R}$ path-connected and Hausdorff. Then there is a unique conformal structure on $\tilde{R}$ making π holomorphic.

Proof. Existence. Let $p \in \tilde{R}$. Since π is a local homeomorphism, choose an open $\tilde{N} \ni p$ mapped homeomorphically by π onto an open $N \ni \pi(p)$. Choose a chart (ϕ, U) on R with $\pi(p) \in U$, and set

$$\tilde{U} = (\pi|_{\tilde{N}})^{-1}(N \cap U), \qquad \tilde{\phi} = \phi \circ \pi|_{\tilde{U}}.$$

Then $\tilde{\phi}$ is a composite of homeomorphisms onto open subsets of $\mathbb{C}$, so $(\tilde{\phi}, \tilde{U})$ is a chart, and these charts cover $\tilde{R}$. For two of them the transition function is

$$\tilde{\phi}_1 \circ \tilde{\phi}_2^{-1} = \phi_1 \circ \pi \circ \pi^{-1} \circ \phi_2^{-1} = \phi_1 \circ \phi_2^{-1},$$

(with the evident restrictions), which is holomorphic since ϕ_1, ϕ_2 came from an atlas on R. So we have an atlas, and by Lemma 4.9 a conformal structure $\tilde{\mathcal{A}}$.

In this structure π is holomorphic: with $\tilde{\phi}$ built from ϕ as above, the local expression of π is

$$\phi \circ \pi \circ \tilde{\phi}^{-1} = \phi \circ \pi \circ \pi^{-1} \circ \phi^{-1} = \mathrm{id},$$

and Lemma 4.15 applies.

Uniqueness. Let $\tilde{\mathcal{B}}$ be any conformal structure on $\tilde{R}$ making π holomorphic. If $(\psi, V) \in \tilde{\mathcal{B}}$ and $(\tilde{\phi}, \tilde{U})$ is one of the charts constructed above, then on the overlap

$$\tilde{\phi} \circ \psi^{-1} = \phi \circ \pi \circ \psi^{-1}$$

is holomorphic, being the local expression of π in charts ψ and ϕ. Likewise $\psi \circ \tilde{\phi}^{-1}$ is holomorphic, since $\pi|_{\tilde{U}}$ has a holomorphic local inverse (its local expression is the identity). Hence every chart of $\tilde{\mathcal{B}}$ is compatible with our atlas, and by maximality $\tilde{\mathcal{B}} = \tilde{\mathcal{A}}$. □

Example 4.22 (The Surface of the Logarithm, Again)

Since $\pi : R \to \mathbb{C}^*$ is a covering map and R is path-connected and Hausdorff, R becomes a Riemann surface with π holomorphic. Moreover $f : R \to \mathbb{C}$ is holomorphic, since locally $f|_{U_n} = f_n \circ \pi$ and f_n is holomorphic.

In fact we can identify R outright. Each $f_n : U_n \to \tilde{V}_{I(n)} = \mathbb{R} + iI(n)$ is a homeomorphism with inverse exp, and these agree on overlaps by construction of $\sim$; so they patch to a bijection $f : R \to \mathbb{C}$, which is holomorphic with holomorphic inverse. Hence

$$R \cong \mathbb{C}.$$

The Riemann surface of the logarithm is just the plane – but presented as a spiral over $\mathbb{C}^*$, with π the map exp in disguise. This is our first hint that the interesting data is not the surface alone but the surface *together with its map to the base.*

Example 4.23 (The Surface of the k-th Root)

Similarly R_k becomes a Riemann surface with π, g holomorphic, and $g : R_k \to \mathbb{C}^*$ is a conformal equivalence, with $\pi = p_k \circ g$. So $R_k \cong \mathbb{C}^*$, again realised as a k-sheeted spiral over $\mathbb{C}^*$.

§4.6 Complex Tori

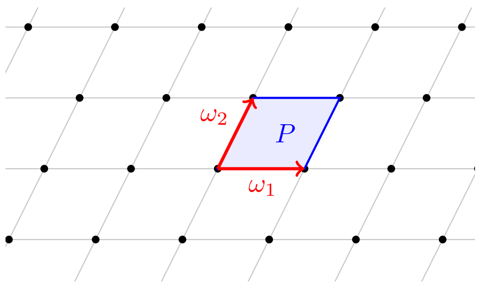
So far our only compact example is $\mathbb{C}_\infty$. Here is a family of others.

Definition 4.24 (Lattice)

Let $\omega_1, \omega_2 \in \mathbb{C}^*$ be linearly independent over $\mathbb{R}$. The subgroup

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \leq (\mathbb{C}, +)$$

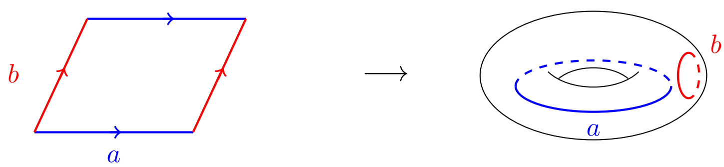
is called a **lattice** in $\mathbb{C}$.



Definition 4.25 (Complex Torus)

The quotient group $T = \mathbb{C}/\Lambda$, with the quotient topology, is called a **complex torus**.

The **fundamental parallelogram** $P = \{s\omega_1 + t\omega_2 : s, t \in [0, 1]\}$ maps onto T, and is injective on its interior, so T is compact (being a continuous image of the compact set P) and indeed $T \cong S^1 \times S^1$ topologically, with the two pairs of opposite edges of P glued.



We must give T a conformal structure. Let $\pi : \mathbb{C} \to T$ be the quotient map.

Proposition 4.26

$\pi : \mathbb{C} \to T$ is a regular covering map, and hence T carries a unique conformal structure making π holomorphic.

Proof. First, Λ is discrete: it is a lattice, so $\mu := \min\{|\lambda| : \lambda \in \Lambda \setminus \{0\}\}$ exists and is positive. (Indeed only finitely many lattice points lie in any bounded set, since $|m\omega_1 + n\omega_2| \geq c(|m| + |n|)$ for a constant $c > 0$ depending on ω_1, ω_2; we prove this in Lemma 11.9.) Fix $0 < \varepsilon < \mu/2$.

π is an open map, since for $U \subseteq \mathbb{C}$ open, $\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in \Lambda}(U + \lambda)$ is open. Given $z_0 \in \mathbb{C}$, put $V = \pi(D(z_0, \varepsilon))$, which is open, and

$$\pi^{-1}(V) = \bigcup_{\lambda \in \Lambda}(D(z_0, \varepsilon) + \lambda) = \coprod_{\lambda \in \Lambda}(D(z_0, \varepsilon) + \lambda),$$

the union being disjoint because two of these discs meeting would give $|\lambda - \lambda'| < 2\varepsilon < \mu$. Each translate maps bijectively (hence homeomorphically, π being open and continuous) onto V. So π is a regular covering map, Λ with the discrete topology indexing the sheets.

T is Hausdorff: given $\pi(z_0) \neq \pi(z_1)$, the sets $\pi(D(z_0, \delta))$ and $\pi(D(z_1, \delta))$ are disjoint open neighbourhoods once 2δ is less than the distance from z_1 to $z_0 + \Lambda$. It is path-connected as a continuous image of $\mathbb{C}$. Now apply Lemma 4.21. $\square$

Explicitly, the charts on T are the maps $(\pi|_{D(z_0,\varepsilon)})^{-1}$ defined on $\pi(D(z_0, \varepsilon))$, and the transition functions are the translations $z \mapsto z + \lambda$, which are certainly holomorphic. This is the cleanest possible atlas, and it is why tori are such a good testing ground.

Remark. All complex tori are homeomorphic, being homeomorphic to $S^1 \times S^1$. They are very far from being all conformally equivalent: one shows on the example sheets that $\mathbb{C}/\Lambda \cong \mathbb{C}/\Lambda'$ if and only if $\Lambda' = a\Lambda$ for some $a \in \mathbb{C}^*$, so that the conformal equivalence classes are parametrised by the modulus $\tau = \omega_2/\omega_1 \in \mathbb{H}$ up to the action of $\mathrm{SL}_2(\mathbb{Z})$. There are uncountably many. So conformal structure is a strictly finer invariant than topology – which is exactly why the subject is interesting.

§5 Functions on Riemann Surfaces

We have a supply of surfaces; now we do analysis on them. Every result in this section is proved the same way: take charts, quote the corresponding theorem in $\mathbb{C}$, and check that the conclusion is chart-independent. What makes the section worth reading is the consequences, which are of a kind that has no analogue in the plane.

§5.1 The Identity Theorem

Theorem 5.1 (Identity Theorem for Riemann Surfaces)
Let $f, g : R \to S$ be holomorphic maps of Riemann surfaces. If the set

$$A = \{p \in R : f(p) = g(p)\}$$

has an accumulation point in R, then $f = g$.

Proof. Let B be the set of accumulation points of A in R; by hypothesis $B \neq \emptyset$. We show B is open and closed in R, which by connectedness gives $B = R$ and hence (as $A \supseteq B$ by continuity) $f = g$.

B is open. Let $p \in B$. Then $f(p) = g(p) =: q$ by continuity. Choose a chart (ψ, V) on S about q and a connected chart (ϕ, U) on R about p with $f(U), g(U) \subseteq V$; this is possible since f, g are continuous. The local expressions $\psi f \phi^{-1}$ and $\psi g \phi^{-1}$ are holomorphic on the domain $\phi(U)$ and agree on a set with an accumulation point (namely $\phi(A \cap U)$, which accumulates at $\phi(p)$). By the identity theorem in $\mathbb{C}$ (Corollary 1.4) they agree on $\phi(U)$, so $f = g$ on U, and every point of U is an accumulation point of A. Hence $U \subseteq B$.

B is closed. Suppose $p \in \bar{B}$. Take U as above; U meets B, so U contains points of A other than p, whence by the argument above $f = g$ on U and $p \in B$. □

Corollary 5.2

A non-constant holomorphic map $f : R \to S$ is nowhere locally constant, and its fibres $f^{-1}(q)$ are discrete.

Proof. If f were constant $= q$ on an open set, apply Theorem 5.1 to f and the constant map q. For the fibres: if $f^{-1}(q)$ had an accumulation point, the same comparison gives f constant. □

§5.2 The Open Mapping Theorem

Theorem 5.3 (Open Mapping Theorem)

Every non-constant holomorphic map $f : R \to S$ of Riemann surfaces is an open map.

Proof. Let $W \subseteq R$ be open and $p \in W$; we find a neighbourhood of $f(p)$ inside $f(W)$. Choose charts (ϕ, U) about p and (ψ, V) about $f(p)$, and shrink so that $U \subseteq W$, U is connected and $f(U) \subseteq V$. By Corollary 5.2 the local expression $F = \psi f \phi^{-1}$ is non-constant on the domain $\phi(U)$, so by the open mapping theorem in $\mathbb{C}$ (Theorem 1.11) $F(\phi(U))$ is open in $\mathbb{C}$. Hence

$$f(U) = \psi^{-1}\left(F(\phi(U))\right)$$

is open in S, being the preimage of an open set under the homeomorphism ψ restricted appropriately. As $f(U) \subseteq f(W)$ contains $f(p)$, we are done. □

Corollary 5.4

Let $f : R \to S$ be a non-constant holomorphic map with R compact. Then f is surjective and S is compact.

Proof. $f(R)$ is open by Theorem 5.3 and compact as a continuous image of a compact set, hence closed since S is Hausdorff. As S is connected and $f(R)$ is a non-empty clopen subset, $f(R) = S$. Then $S = f(R)$ is compact. □

Corollary 5.5

Every holomorphic function $f : R \to \mathbb{C}$ on a compact Riemann surface R is constant.

Proof. If f were non-constant then $\mathbb{C}$ would be compact by Corollary 5.4. It is not. □

This is a striking statement, and it deserves to be dwelt on: it is a vast generalisation of Liouville's theorem. On a compact surface there is *no* interesting holomorphic function at all. All the analysis on a compact Riemann surface must therefore be carried by maps to other compact targets – to $\mathbb{C}_\infty$, giving meromorphic functions, or to other surfaces. This single fact shapes the rest of the course.

Theorem 5.6 (Maximum Modulus Principle)

Let $f : R \to \mathbb{C}$ be holomorphic on a Riemann surface. If $|f|$ attains a local maximum at some $p \in R$, then f is constant.

Proof. Suppose f is non-constant, so $f(R)$ is open by Theorem 5.3. Let W be a neighbourhood of p with $|f(q)| \leq |f(p)|$ for $q \in W$. Then $f(W)$ is an open neighbourhood of $f(p)$ contained in the closed disc $\overline{D}(0, |f(p)|)$; but every neighbourhood of $f(p)$ contains points of modulus greater than $|f(p)|$ (take $f(p)(1+\epsilon)$ if $f(p) \neq 0$, or any small non-zero point if $f(p) = 0$). This is a contradiction. □

Combining these gives an alternative proof of Corollary 5.5: on a compact R the continuous function $|f|$ attains a maximum, so f is constant.

§5.3 Harmonic Functions

The open mapping theorem tells us that a non-constant holomorphic function cannot be real-valued, since $\mathbb{R} \subseteq \mathbb{C}$ has empty interior. Real-valued functions on Riemann surfaces are therefore never holomorphic; the right class is the harmonic ones.

Definition 5.7 (Harmonic Function)

A smooth function $u : D \to \mathbb{R}$ on a domain $D \subseteq \mathbb{C}$ is **harmonic** if

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Lemma 5.8

Let $D \subseteq \mathbb{C}$ be a disc. A function $u : D \to \mathbb{R}$ is harmonic if and only if $u = \operatorname{Re} f$ for some holomorphic f on D.

Proof. If $f = u + iv$ is holomorphic then the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$ give $u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$. The converse – constructing a harmonic conjugate v on a disc by integrating $-u_y \, dx + u_x \, dy$, which is closed by harmonicity and hence exact on a simply connected domain – was an exercise in IB Complex Analysis. □

Note the appearance of simple connectedness: the converse fails on an annulus, as $u = \log|z|$ shows.

Definition 5.9

A function $u : R \to \mathbb{R}$ on a Riemann surface is **harmonic** if $u \circ \phi^{-1}$ is harmonic for every chart (ϕ, U).

For this to be a sensible definition we need harmonicity to be preserved by holomorphic changes of coordinate, which it is:

Lemma 5.10

A function $u : R \to \mathbb{R}$ is harmonic if and only if each $p \in R$ has *some* chart (ϕ, U) with $p \in U$ and $u \circ \phi^{-1}$ harmonic.

Proof. One direction is trivial. Conversely, suppose the condition holds and let (ψ, V) be any chart, $p \in V$. Choose (ϕ, U) as hypothesised with $p \in U$; shrinking, we may assume $\phi(U)$ is a disc, so $u \circ \phi^{-1} = \operatorname{Re} f$ for some holomorphic f by Lemma 5.8. Then near $\psi(p)$,

$$u \circ \psi^{-1} = (u \circ \phi^{-1}) \circ (\phi \circ \psi^{-1}) = \operatorname{Re}\left(f \circ (\phi \circ \psi^{-1})\right),$$

the real part of a holomorphic function, hence harmonic. □

Theorem 5.11 (Open Mapping and Identity Theorems for Harmonic Functions)

Let u, v be harmonic on a Riemann surface R.

(i) If $\{p : u(p) = v(p)\}$ has non-empty interior then $u = v$.

(ii) If u is non-constant then $u : R \to \mathbb{R}$ is an open map.

Consequently, every harmonic function on a compact Riemann surface is constant.

Proof. Locally $u = \operatorname{Re} f$ with f holomorphic. For (i), if $u = v$ on an open set then locally $\operatorname{Re}(f - g) = 0$, so $f - g$ is constant (its image lies in $i\mathbb{R}$, which has empty interior, so it cannot be open), and a clopen argument as in Theorem 5.1 propagates this. For (ii), openness of u follows from openness of f on any chart where f is non-constant, since $\operatorname{Re} : \mathbb{C} \to \mathbb{R}$ is open. The last statement follows since $u(R)$ would be an open, compact – hence clopen and non-empty – subset of the connected space $\mathbb{R}$, forcing $u(R) = \mathbb{R}$, which is not compact. □

§5.4 Meromorphic Functions

Since holomorphic functions to $\mathbb{C}$ are useless on compact surfaces, we enlarge the target to the compact surface $\mathbb{C}_\infty$.

Definition 5.12 (Meromorphic Function)

A **meromorphic function** on a Riemann surface R is a holomorphic map $f : R \to \mathbb{C}_\infty$ which is not identically ∞.

Note that $f^{-1}(\infty)$ is then discrete by Corollary 5.2, and closed since $\{\infty\}$ is; so a meromorphic function is a holomorphic function on the complement of a discrete closed set. The next result confirms that on a plane domain this agrees with the classical definition.

Proposition 5.13

Let $D \subseteq \mathbb{C}$ be a domain. A map $f : D \to \mathbb{C}_\infty$ is meromorphic if and only if there is a discrete closed $A \subseteq D$ such that $f|_{D \setminus A}$ is holomorphic with values in $\mathbb{C}$, and f has a pole at each point of A.

Proof. ($\Rightarrow$) Let $A = f^{-1}(\infty)$, which is discrete and closed as noted, and certainly f is holomorphic $D \setminus A \to \mathbb{C}$. Fix $a \in A$ and work in the chart $w \mapsto 1/w$ about ∞. The local expression $z \mapsto 1/f(z)$ is holomorphic near a and vanishes at a; it is not identically zero (else $f \equiv \infty$), so by Proposition 1.3 we may write $1/f(z) = (z-a)^m g(z)$ with $m \geq 1$ and $g(a) \neq 0$. Shrinking so that g has no zero, $f(z) = (z-a)^{-m}/g(z)$, which has a pole of order m at a.

($\Leftarrow$) f is certainly holomorphic as a map into $\mathbb{C}_\infty$ on $D \setminus A$. At $a \in A$ with a pole of order m we may write $f(z) = (z-a)^{-m}h(z)$ with h holomorphic and $h(a) \neq 0$; then $1/f(z) = (z-a)^m/h(z)$ is holomorphic near a (after shrinking so h does not vanish), and f is continuous into $\mathbb{C}_\infty$ at a since $|f| \to \infty$. So the local expression in the chart at ∞ is holomorphic, and Lemma 4.15 applies. $\square$

Example 5.14

The map $\mathbb{C}_\infty \to \mathbb{C}_\infty$ given by a rational function $p(z)/q(z)$ (in lowest terms, with the evident values at the zeros of q and at ∞) is meromorphic. We shall see in Section 11 that these are the *only* meromorphic functions on $\mathbb{C}_\infty$.

Example 5.15

On a complex torus $T = \mathbb{C}/\Lambda$, a meromorphic function pulls back along $\pi : \mathbb{C} \to T$ to a meromorphic function on $\mathbb{C}$ invariant under Λ – that is, to a doubly periodic, or **elliptic**, function. Conversely every such function descends. Corollary 5.5 then says: a doubly periodic *holomorphic* function on $\mathbb{C}$ is constant, which is Liouville's theorem applied to the (bounded, by compactness of P) function. Section 11 is devoted to the non-constant ones.

§6 Covering Spaces and Monodromy

To organise analytic continuation we need the topology of covering spaces. Most of this was proved in II Algebraic Topology, and we shall be brisk, but the path-lifting and homotopy-lifting statements are so central that it is worth seeing the proofs again in the form we need.

§6.1 Lifting

Definition 6.1 (Lift)
Let $\pi : \tilde{X} \to X$ be a covering map and $\gamma : [0,1] \to X$ a path. A **lift** of γ is a path $\tilde{\gamma} : [0,1] \to \tilde{X}$ with $\pi \circ \tilde{\gamma} = \gamma$.

Example 6.2
For $\exp : \mathbb{C} \to \mathbb{C}^*$ and $\gamma(t) = e^{2\pi i t}$, both $t \mapsto 2\pi i t$ and $t \mapsto 2\pi i + 2\pi i t$ are lifts. They differ, but they start at different points – and that is the general phenomenon.

Proposition 6.3 (Uniqueness of Lifts)
Let $\pi : \tilde{X} \to X$ be a covering map and $\tilde{\gamma}_1, \tilde{\gamma}_2$ lifts of the same path γ. If $\tilde{\gamma}_1(0) = \tilde{\gamma}_2(0)$ then $\tilde{\gamma}_1 = \tilde{\gamma}_2$.

Proof. Let $I = \{t \in [0,1] : \tilde{\gamma}_1(t) = \tilde{\gamma}_2(t)\}$, which is non-empty and closed (as $\tilde{X}$ is Hausdorff and the $\tilde{\gamma}_i$ continuous). We show I is open, so that $I = [0,1]$ by connectedness.

Let $t \in I$ and put $\tilde{x} = \tilde{\gamma}_1(t) = \tilde{\gamma}_2(t)$. As π is a local homeomorphism there is an open $\tilde{N} \ni \tilde{x}$ with $\pi|_{\tilde{N}}$ a homeomorphism onto an open N. By continuity there is $\delta > 0$ with $\tilde{\gamma}_i((t-\delta, t+\delta)) \subseteq \tilde{N}$ for $i = 1, 2$. For such s,

$$\tilde{\gamma}_1(s) = (\pi|_{\tilde{N}})^{-1}(\gamma(s)) = \tilde{\gamma}_2(s),$$

so $(t-\delta, t+\delta) \cap [0,1] \subseteq I$. □

Existence, however, requires regularity.

Example 6.4
Let $\tilde{X} = \mathbb{R} + i(-\pi, 2\pi)$ and $\pi = \exp|_{\tilde{X}} : \tilde{X} \to \mathbb{C}^*$. This is a covering map (a local homeomorphism onto $\mathbb{C}^*$), but the loop $t \mapsto e^{2\pi i t}$ cannot be lifted starting at 0: the lift would have to be $t \mapsto 2\pi i t$, which leaves $\tilde{X}$ at $t = 1$. The map π is not regular.

Proposition 6.5 (Path Lifting)
Let $\pi : \tilde{X} \to X$ be a regular covering map, $\gamma : [0,1] \to X$ a path and $\tilde{x} \in \tilde{X}$ with $\pi(\tilde{x}) = \gamma(0)$. Then there is a unique lift $\tilde{\gamma}$ of γ with $\tilde{\gamma}(0) = \tilde{x}$.

Proof. Uniqueness is Proposition 6.3. For existence let

$$I = \{t \in [0,1] : \gamma|_{[0,t]} \text{ lifts to some } \tilde{\gamma} \text{ with } \tilde{\gamma}(0) = \tilde{x}\},$$

which contains 0. Note that by uniqueness, lifts on $[0,t]$ for varying t are compatible, so I is an interval and the lifts patch.

I is open in $[0,1]$. Let $\tau \in I$ with lift $\tilde{\gamma}$ on $[0,\tau]$. Choose an evenly covered $U \ni \gamma(\tau)$, so $\pi^{-1}(U) = \coprod_\delta U_\delta$, and let U_{δ_0} be the sheet containing $\tilde{\gamma}(\tau)$. Choose $\varepsilon > 0$ with

$\gamma((\tau - \varepsilon, \tau + \varepsilon)) \subseteq U$; then setting $\tilde{\gamma}(t) = (\pi|_{U_{\delta_0}})^{-1}(\gamma(t))$ for $t \in [\tau, \tau + \varepsilon)$ extends the lift.

I is closed. Let $\tau = \sup I$ and choose an evenly covered $U \ni \gamma(\tau)$ with sheets U_δ. Pick $\varepsilon > 0$ with $\gamma((\tau - \varepsilon, \tau]) \subseteq U$ and choose $t \in I$ with $t > \tau - \varepsilon$. The connected set $\tilde{\gamma}((\tau - \varepsilon, t])$ lies in $\pi^{-1}(U)$, hence in a single sheet U_{δ_0}, and we may define $\tilde{\gamma}(s) = (\pi|_{U_{\delta_0}})^{-1}(\gamma(s))$ for $s \in (\tau - \varepsilon, \tau]$, which agrees with the existing lift and extends it continuously to τ. So $\tau \in I$.

Hence I is a non-empty clopen subset of $[0, 1]$, so $I = [0, 1]$. □

Theorem 6.6 (Homotopy Lifting, or the Monodromy Theorem)
Let $\pi : \tilde{X} \to X$ be a regular covering map and let $\alpha \simeq \beta$ be homotopic paths in X (rel endpoints). Let $\tilde{\alpha}, \tilde{\beta}$ be their lifts with $\tilde{\alpha}(0) = \tilde{\beta}(0)$. Then $\tilde{\alpha} \simeq \tilde{\beta}$; in particular $\tilde{\alpha}(1) = \tilde{\beta}(1)$.

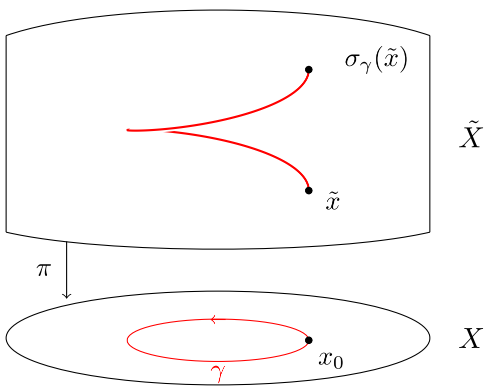
Proof. This is the homotopy lifting property, proved in II Algebraic Topology: one lifts the homotopy $H : [0, 1]^2 \to X$ by subdividing the square into small squares each mapping into an evenly covered set, and lifting them one at a time, using uniqueness of lifts to see the pieces agree. The resulting $\tilde{H}$ is a homotopy from $\tilde{\alpha}$ to $\tilde{\beta}$; since H is constant on the two vertical edges, $\tilde{H}$ is too (a lift of a constant path is constant, by uniqueness), so the homotopy is rel endpoints. □

The name 'monodromy theorem' attached to this statement is traditional and slightly unfortunate, since we shall also use it for the classical statement about analytic continuation (Theorem 7.12). The latter is a corollary of the former, and the two are best regarded as the same theorem stated on either side of the dictionary we are about to build.

§6.2 The Monodromy Group

Let $\pi : \tilde{X} \to X$ be a regular covering map and fix a basepoint $x_0 \in X$. For a loop γ at x_0 and $\tilde{x} \in \pi^{-1}(x_0)$, let $\tilde{\gamma}_{\tilde{x}}$ be the lift starting at $\tilde{x}$; since $\pi(\tilde{\gamma}_{\tilde{x}}(1)) = \gamma(1) = x_0$, the endpoint again lies in the fibre.

Definition 6.7 (Monodromy Action)
Define $\sigma_\gamma : \pi^{-1}(x_0) \to \pi^{-1}(x_0)$ by $\sigma_\gamma(\tilde{x}) = \tilde{\gamma}_{\tilde{x}}(1)$.



Proposition 6.8

The assignment $\gamma \mapsto \sigma_\gamma$ satisfies:

(i) $\sigma_c = \mathrm{id}$ for the constant loop c;

(ii) $\sigma_{\bar\gamma} = \sigma_\gamma^{-1}$, where $\bar\gamma(t) = \gamma(1-t)$; in particular each σ_γ is a bijection;

(iii) $\sigma_{\alpha\cdot\beta} = \sigma_\beta \circ \sigma_\alpha$;

(iv) if $\alpha \simeq \beta$ then $\sigma_\alpha = \sigma_\beta$.

Hence $\{\sigma_\gamma\}$ is a subgroup of $\mathrm{Sym}(\pi^{-1}(x_0))$, and there is a well-defined (right) action of $\pi_1(X, x_0)$ on the fibre.

Proof. (i) is clear. For (iii), if $\tilde\alpha_{\tilde x}$ is the lift of α at $\tilde x$, then the concatenation $\tilde\alpha_{\tilde x} \cdot \tilde\beta_{\tilde\alpha_{\tilde x}(1)}$ is a lift of $\alpha \cdot \beta$ starting at $\tilde x$, so by uniqueness it is *the* lift, and its endpoint is $\sigma_\beta(\sigma_\alpha(\tilde x))$. Then (ii) follows from (i) and (iii). Finally (iv) is Theorem 6.6. ☐

Definition 6.9 (Monodromy Group)

The subgroup $\{\sigma_\gamma : \gamma \text{ a loop at } x_0\} \le \mathrm{Sym}(\pi^{-1}(x_0))$ is the **monodromy group** of the covering π.

Up to isomorphism it does not depend on the basepoint, since a path from x_0 to x_1 conjugates the two actions.

Example 6.10

Take $p_k : \mathbb{C}^* \to \mathbb{C}^*$, $z \mapsto z^k$, with basepoint 1. The fibre is the set of k-th roots of unity ζ^n, $\zeta = e^{2\pi i/k}$. Let $\gamma(t) = e^{2\pi i t}$ be the standard loop; its lift starting at ζ^n is $t \mapsto \zeta^n e^{2\pi i t/k}$, which ends at ζ^{n+1}. So σ_γ is the k-cycle $\zeta^n \mapsto \zeta^{n+1}$. Since $\pi_1(\mathbb{C}^*) \cong \mathbb{Z}$ is generated by $[\gamma]$, the monodromy group is cyclic of order k.

This is the precise sense in which '$\sqrt[k]{z}$ has k branches which are permuted cyclically as z winds once round 0'.

Example 6.11

For $\exp : \mathbb{C} \to \mathbb{C}^*$ with basepoint 1, the fibre is $2\pi i\mathbb{Z}$, the loop γ lifts to $t \mapsto 2\pi i t + 2\pi i n$ starting at $2\pi i n$, and σ_γ is the shift $2\pi i n \mapsto 2\pi i(n+1)$. The monodromy group is infinite cyclic, matching the infinitely many branches of log.

§7 The Space of Germs

We now build, once and for all, the Riemann surface on which a complete analytic function becomes single-valued. The construction of Section 3 was ad hoc: we chose particular sectors and glued them by hand. The trick is to stop choosing. Take *all* function elements at once, remember only their local behaviour, and let the topology sort things out.

§7.1 Germs

Fix a domain $D \subseteq \mathbb{C}$.

Definition 7.1 (Germ)
Let (f, U) and (g, V) be function elements on D and let $z \in U \cap V$. Write $(f, U) \equiv_z (g, V)$ if $f = g$ on some neighbourhood of z. This is an equivalence relation on the function elements whose domain contains z; the equivalence class of (f, U) is the **germ** of f at z, written $[f]_z$.

So $[f]_z = [g]_w$ if and only if $z = w$ and f, g agree near z. A germ at z is exactly the same data as a convergent power series centred at z; but the geometric language will be more useful.

Definition 7.2 (The Space of Germs)
The **space of germs** over D is

$$\mathcal{G} = \mathcal{G}_D = \{[f]_z : z \in D,\ (f, U) \text{ a function element on } D \text{ with } z \in U\}.$$

We topologise $\mathcal{G}$ so that germs of the same function at nearby points are nearby. For a function element (f, U) write

$$[f]_U = \{[f]_z : z \in U\} \subseteq \mathcal{G},$$

a 'sheet' of germs.

Lemma 7.3
The collection $\{[f]_U\}$, over all function elements (f, U) on D, is a basis for a topology on $\mathcal{G}$.

Proof. The sets cover $\mathcal{G}$ by definition. Suppose $[h]_z \in [f]_U \cap [g]_V$. Then h agrees with f near z and with g near z, so on some open $W \ni z$ with $W \subseteq U \cap V$ we have $h = f = g$, whence $[h]_W \subseteq [f]_U \cap [g]_V$ and $[h]_z \in [h]_W$. □

Lemma 7.4
$\mathcal{G}$ is Hausdorff.

Proof. Let $[f]_z \neq [g]_w$. If $z \neq w$, choose disjoint neighbourhoods $U \ni z$, $V \ni w$ inside the domains of f and g; then $[f]_U$ and $[g]_V$ are disjoint, since germs in them sit over disjoint sets of points.

If $z = w$, choose a *connected* open $U \ni z$ contained in both domains, so that (f, U) and (g, U) represent $[f]_z, [g]_z$. Suppose $[h]_y \in [f]_U \cap [g]_U$. Then h agrees with f near y and with g near y, so $f = g$ on a neighbourhood of y, and hence on all of U by the identity theorem (U being connected). But then $[f]_z = [g]_z$, a contradiction. So $[f]_U \cap [g]_U = \emptyset$. □

Definition 7.5

The **forgetful map** $\pi : \mathcal{G} \to D$ is $\pi([f]_z) = z$.

Lemma 7.6

For each connected component G of $\mathcal{G}$, the restriction $\pi : G \to D$ is a covering map onto its image, which is open in D. In particular π is a local homeomorphism.

Proof. π is continuous: for $U \subseteq D$ open, $\pi^{-1}(U) = \bigcup [f]_V$ over all function elements (f, V) with $V \subseteq U$, which is open. On a basic open set, $\pi|_{[f]_U} : [f]_U \to U$ is a bijection with inverse $z \mapsto [f]_z$, and this inverse is continuous because the preimage of a basic set $[g]_V$ is the open set $\{z \in U \cap V : f = g \text{ near } z\}$. So $\pi|_{[f]_U}$ is a homeomorphism onto the open set U, and π is a local homeomorphism. Its image is therefore open, and each component G of $\mathcal{G}$ is path-connected (being locally homeomorphic to open subsets of $\mathbb{C}$, hence locally path-connected) and Hausdorff by Lemma 7.4. □

Remark. Note that $\pi : G \to D$ need not be *regular*. For instance, take $D = \mathbb{C}$ and the component containing the germs of $z \mapsto 1/(1-z)$: continuation is blocked at $z = 1$, and the fibre over 1 can be empty while nearby fibres are not. This is exactly the phenomenon of a function element which cannot be continued along every path.

By Lemma 4.21, each component G of $\mathcal{G}$ inherits a unique conformal structure making π holomorphic. Explicitly, the charts are the maps $\pi|_{[f]_U}$ on the sets $[f]_U$, and the transition functions are identities. And there is a canonical holomorphic function on it:

Definition 7.7 (Evaluation Map)

The **evaluation map** $E : \mathcal{G} \to \mathbb{C}$ is $E([f]_z) = f(z)$.

This is well defined, since germs which agree near z certainly agree at z. In the chart $\pi|_{[f]_U}$ its local expression is

$$E \circ (\pi|_{[f]_U})^{-1}(z) = E([f]_z) = f(z),$$

which is holomorphic. So E is a holomorphic function on $\mathcal{G}$.

This is the whole point. The multi-valued function has become the pair (π, E): π remembers which point of D we are over, and E is a genuine single-valued holomorphic function.

§7.2 Analytic Continuation as Lifting

Now we cash in.

Theorem 7.8

Let $(f, U), (g, V)$ be function elements on a domain D and let $\gamma : [0, 1] \to D$ be a path with $\gamma(0) \in U$, $\gamma(1) \in V$. Then

$$(f, U) \approx_\gamma (g, V) \iff \gamma \text{ lifts along } \pi \text{ to a path in } \mathcal{G} \text{ from } [f]_{\gamma(0)} \text{ to } [g]_{\gamma(1)}.$$

Proof. ($\Rightarrow$) Suppose we have a chain $(f, U) = (f_1, U_1) \sim \cdots \sim (f_n, U_n) = (g, V)$ and a dissection $0 = t_0 < \cdots < t_n = 1$ with $\gamma([t_{i-1}, t_i]) \subseteq U_i$. Define

$$\tilde{\gamma}(t) = [f_i]_{\gamma(t)}, \qquad t \in [t_{i-1}, t_i].$$

This is well defined at the joins: at $t = t_i$ we need $[f_i]_{\gamma(t_i)} = [f_{i+1}]_{\gamma(t_i)}$, which holds since $f_i = f_{i+1}$ on $U_i \cap U_{i+1}$, an open set containing $\gamma(t_i)$. On each $[t_{i-1}, t_i]$ we have $\tilde{\gamma} = (\pi|_{[f_i]_{U_i}})^{-1} \circ \gamma$, which is continuous; by the gluing lemma $\tilde{\gamma}$ is continuous. Clearly $\pi \circ \tilde{\gamma} = \gamma$ and the endpoints are as claimed.

($\Leftarrow$) Suppose $\tilde{\gamma}$ is such a lift. Every point $\tilde{\gamma}(t)$ has a basic neighbourhood $[f_t]_{U_t}$ with U_t a *disc*. By compactness of $[0, 1]$ and the Lebesgue number lemma, there is a dissection $0 = t_0 < \cdots < t_n = 1$ and function elements (f_i, U_i) with U_i a disc and $\tilde{\gamma}([t_{i-1}, t_i]) \subseteq [f_i]_{U_i}$. Applying π gives $\gamma([t_{i-1}, t_i]) \subseteq U_i$. Moreover for each i,

$$[f_i]_{\gamma(t_i)} = \tilde{\gamma}(t_i) = [f_{i+1}]_{\gamma(t_i)},$$

so $f_i = f_{i+1}$ near $\gamma(t_i) \in U_i \cap U_{i+1}$. Since U_i, U_{i+1} are discs, $U_i \cap U_{i+1}$ is convex hence connected, and the identity theorem gives $f_i = f_{i+1}$ on $U_i \cap U_{i+1}$, i.e. $(f_i, U_i) \sim (f_{i+1}, U_{i+1})$. Finally $[f_1]_{\gamma(0)} = [f]_{\gamma(0)}$ and $[f_n]_{\gamma(1)} = [g]_{\gamma(1)}$, and the identity theorem on the connected sets $U_1 \cap U$ and $U_n \cap V$ upgrades these germ equalities to $(f, U) \sim (f_1, U_1)$ and $(f_n, U_n) \sim (g, V)$. □

This is the theorem that makes the space of germs worth building: *analytic continuation along a path is path lifting.* Every fact about continuation is now a fact about covering spaces.

Corollary 7.9

Let $\mathcal{F}$ be a complete analytic function on a domain D. Then

$$G_{\mathcal{F}} = \bigcup_{(f,U) \in \mathcal{F}} [f]_U$$

is a path component of $\mathcal{G}$.

Proof. Two germs lie in the same path component precisely when there is a path in $\mathcal{G}$ between them, which by Theorem 7.8 happens precisely when the corresponding function elements are analytic continuations of one another. □

Definition 7.10

$G_{\mathcal{F}}$, with the conformal structure of Lemma 4.21, is **the Riemann surface of the complete analytic function** $\mathcal{F}$, equipped with $\pi : G_{\mathcal{F}} \to D$ and $E : G_{\mathcal{F}} \to \mathbb{C}$.

Example 7.11

Take $D = \mathbb{C}^*$ and $\mathcal{F}$ the complete analytic function containing the elements $F_n = (f_n, U_n)$ of Section 2.2. Then $G_{\mathcal{F}}$ is precisely the surface R we built by hand, and E is the function f. Indeed $\Phi : R \to G_{\mathcal{F}}$, $[z] \mapsto [f_n]_z$ for $z \in U_n$, is a well-defined homeomorphism commuting with the projections.

§7.3 The Classical Monodromy Theorem

Theorem 7.12 (Classical Monodromy Theorem)

Let $D \subseteq \mathbb{C}$ be a domain and (f, U) a function element on D which can be continued along *every* path in D starting in U. Suppose α, β are homotopic paths in D (rel endpoints) from a point of U to a point of V, and

$$(f, U) \approx_\alpha (g_1, V), \qquad (f, U) \approx_\beta (g_2, V).$$

Then $g_1 = g_2$ on V (assuming V connected).

Proof. Let G be the path component of $\mathcal{G}$ containing $[f]_{\alpha(0)}$. The hypothesis says exactly that every path in D starting at $\alpha(0)$ lifts to G starting at $[f]_{\alpha(0)}$, from which one deduces that $\pi : G \to D$ is a regular covering map: given $z \in D$, take a disc $B \ni z$ in D; each germ in $\pi^{-1}(z)$ extends over B (continue along radii and use the identity theorem, or note that by Theorem 7.8 the radial paths lift), giving a sheet of $\pi^{-1}(B)$ over B, and these sheets are disjoint and exhaust $\pi^{-1}(B)$.

Now let $\tilde{\alpha}, \tilde{\beta}$ be the lifts of α, β starting at $[f]_{\alpha(0)}$, which exist by Theorem 7.8 and end at $[g_1]_{\alpha(1)}$, $[g_2]_{\beta(1)}$ respectively. Since $\alpha \simeq \beta$, Theorem 6.6 gives $\tilde{\alpha}(1) = \tilde{\beta}(1)$, i.e.

$$[g_1]_{\alpha(1)} = [g_2]_{\alpha(1)}.$$

So $g_1 = g_2$ near $\alpha(1)$, hence on V by the identity theorem. $\square$

Corollary 7.13

Let D be a simply connected domain and (f, U) a function element on D which can be continued along every path in D starting in U. Then there is a (unique) holomorphic $F : D \to \mathbb{C}$ with $F|_U = f$.

Proof. For $z \in D$ choose a path γ from a point of U to z and define $F(z)$ to be the value at z of the continuation along γ. Any two such paths are homotopic rel endpoints, so Theorem 7.12 says F is well defined; and it is holomorphic since near each point it agrees with a function element. $\square$

This explains, at last, the branch cuts of a first course. There is no holomorphic logarithm on $\mathbb{C}^*$, because $\mathbb{C}^*$ is not simply connected and continuation once round the origin changes the branch. Slitting the plane makes it simply connected, so a branch exists; but the slit is a choice, and the honest object is the covering $\pi : R \to \mathbb{C}^*$ which forgets no branch at all.

§8 Gluing Riemann Surfaces

The surfaces we have built so far are all non-compact – and by Corollary 5.5 it is compact surfaces where the interesting rigidity lives. The missing operation is compactification: adding the points that continuation cannot reach. The general mechanism is gluing.

§8.1 The Gluing Construction

Definition 8.1 (Gluing)

Let X, Y be topological spaces with subspaces $X' \subseteq X$, $Y' \subseteq Y$ and a homeomorphism $\Phi : X' \to Y'$. The result of **gluing X to Y along Φ** is

$$X \cup_\Phi Y = (X \sqcup Y)/\sim,$$

where $\sim$ is generated by $x \sim \Phi(x)$ for $x \in X'$, with the quotient topology.

Proposition 8.2

Let R_1, R_2 be Riemann surfaces and $S_j \subseteq R_j$ non-empty open connected subsets, and let $\Phi : S_1 \to S_2$ be a conformal equivalence. Then $R = R_1 \cup_\Phi R_2$ carries a unique conformal structure making the inclusions $i_j : R_j \to R$ holomorphic (indeed conformal onto their open images). If R is Hausdorff, then R is a Riemann surface.

Proof. Each i_j is an open injection, so we may regard R_1, R_2 as open subsets of R covering it. Take as atlas all charts of the form $(\phi_j \circ i_j^{-1}, i_j(U_j))$ where (ϕ_j, U_j) is a chart of R_j. Two such charts coming from the same R_j have a transition function which is a transition function of R_j. A chart from R_1 and one from R_2 overlap only inside $i_1(S_1) = i_2(S_2)$, where the transition function is

$$\phi_2 \circ i_2^{-1} \circ i_1 \circ \phi_1^{-1} = \phi_2 \circ \Phi \circ \phi_1^{-1},$$

holomorphic since Φ is. So we have an atlas; extend it by Lemma 4.9. The local expression of i_j in these charts is the identity, so i_j is holomorphic. Uniqueness follows as in Lemma 4.21: any structure making the i_j holomorphic and locally invertible must contain, hence by maximality equal, this one.

Finally R is connected: it is the union of the connected sets $i_1(R_1)$ and $i_2(R_2)$, which meet. □

The Hausdorff hypothesis is not automatic, and one must always check it.

Example 8.3 (The Line with Two Origins)

Take $R_1 = R_2 = \mathbb{C}$ and $S_1 = S_2 = \mathbb{C}^*$ with $\Phi = \mathrm{id}$. Then R is the plane with a doubled origin: the two copies of 0 have no disjoint neighbourhoods, since every punctured neighbourhood of either is identified with that of the other. So R is not Hausdorff and is not a Riemann surface.

Example 8.4 (The Sphere by Gluing)

Take $R_1 = R_2 = \mathbb{C}$, $S_1 = S_2 = \mathbb{C}^*$, and $\Phi(z) = 1/z$, a conformal equivalence $\mathbb{C}^* \to \mathbb{C}^*$. The glued space R is Hausdorff: given the two 'extra' points $i_1(0)$ and $i_2(0)$, the sets $i_1(D(0,1))$ and $i_2(D(0,1))$ are disjoint open neighbourhoods, since Φ maps $D(0,1) \setminus \{0\}$ to the exterior of the closed unit disc. Other pairs of points are separated inside a single copy of $\mathbb{C}$. It is compact, being the union of the two compact sets $i_j(\overline{D}(0,1))$. And of course $R \cong \mathbb{C}_\infty$: the map sending $i_1(z) \mapsto z$ and $i_2(w) \mapsto 1/w$ is a conformal equivalence, since the two charts of $\mathbb{C}_\infty$ are related by exactly Φ.

This is the model for everything that follows: to compactify a surface, find a second surface containing the missing points and glue along the common part.

§8.2 The Surface of $w^2 = z^3 - z$

Let us carry out a substantial example by hand. Set

$$h(z) = z^3 - z = z(z-1)(z+1),$$

and consider the multi-valued function $w = \sqrt{h(z)}$. Near any point where h does not vanish we can choose a square root holomorphically; the trouble is at $z = 0, \pm 1$, and (as we shall see) at ∞.

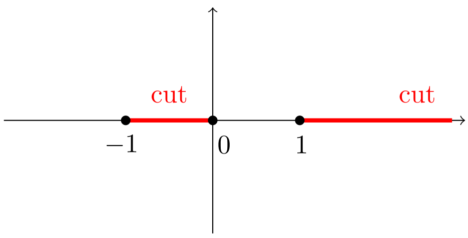
Step 1: a cut domain. Let

$$D = \mathbb{C} \setminus ([-1, 0] \cup [1, \infty)).$$

We construct branches of $\sqrt{h}$ on D. Fix $z_0 \in D$ and w_0 with $w_0^2 = h(z_0)$, and define, for $z \in D$,

$$g(z) = w_0 \exp\left(\frac{1}{2}\int_\gamma \frac{h'(\zeta)}{h(\zeta)}\,\mathrm{d}\zeta\right)$$

where γ is any path in D from z_0 to z.



Why this is well defined. If γ is a loop in D, the argument principle gives

$$\frac{1}{2\pi i}\int_\gamma \frac{h'}{h}\,\mathrm{d}\zeta = n(\gamma, 0) + n(\gamma, 1) + n(\gamma, -1),$$

the sum of winding numbers about the zeros of h (there are no poles). Now γ lies in D, so it cannot wind around 1 at all – the cut $[1, \infty)$ blocks it – giving $n(\gamma, 1) = 0$; and it must wind the same number of times about 0 and about -1, since the cut $[-1, 0]$ joins them and γ cannot separate them. Hence the integral is $2\pi i \cdot 2n(\gamma, 0)$, an even multiple of $2\pi i$, and so

$$\exp\left(\tfrac{1}{2}\int_\gamma \frac{h'}{h}\right) = \exp(2\pi i\, n(\gamma, 0)) = 1.$$

Therefore g does not depend on γ, and it is holomorphic (locally it is $w_0 \exp(\frac{1}{2}L)$ for a local branch L of $\log h$), with

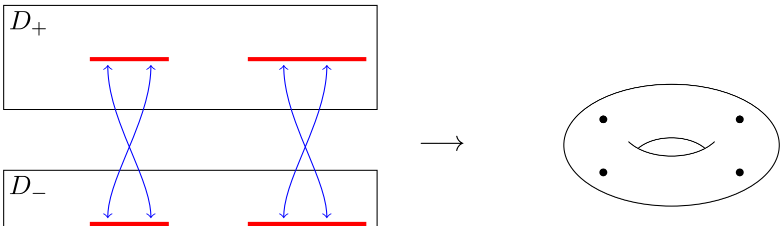
$$g(z)^2 = w_0^2 \exp\left(\int_\gamma \frac{h'}{h}\right) = h(z_0) \cdot \frac{h(z)}{h(z_0)} = h(z).$$

The two choices of w_0 give two branches $g_+ = -g_-$. Write D_+, D_- for two copies of D carrying g_+, g_- respectively.

Step 2: gluing along the cuts. As a subset of the sphere, D is $\mathbb{C}_\infty$ minus two closed arcs, hence homeomorphic to an open cylinder $S^1 \times \mathbb{R}$. For z_0 in the interior of a cut, and writing $z \to z_0^\pm$ for approach from the upper and lower half planes,

$$\lim_{z \to z_0^-} g_+(z) = \lim_{z \to z_0^+} g_-(z), \qquad \lim_{z \to z_0^+} g_+(z) = \lim_{z \to z_0^-} g_-(z).$$

(Crossing a cut flips the sign of the square root, because the argument of exactly one of the factors of h jumps by 2π.) So we glue the upper side of each cut in D_+ to the lower side of the corresponding cut in D_-, and vice versa.



torus, 4 points removed

Each $D_\pm$ is a cylinder; gluing two cylinders along two pairs of slits produces a torus with the four points $0, \pm 1, \infty$ removed. (Think of each cylinder as a sphere with two slits; gluing crosswise along two slits is exactly the standard construction of a genus-1 surface from two spheres with two handles' worth of identification.) We shall verify this by computation in Section 10, and the reader should not feel obliged to believe the picture until then.

Call the resulting Riemann surface R. The two branches assemble into a holomorphic $g : R \to \mathbb{C}$, and the two inclusions $D_\pm \hookrightarrow \mathbb{C} \setminus \{0, \pm 1\}$ assemble into a holomorphic covering map $\pi : R \to \mathbb{C} \setminus \{0, \pm 1\}$, with

$$g(p)^2 = \pi(p)^3 - \pi(p) \qquad \text{for all } p \in R.$$

§8.3 Compactifying the Cubic

The surface R above is missing four points. To restore them, we use the graph description. Let

$$R_1 = \{(z, w) \in \mathbb{C}^2 : w^2 = z^3 - z\}.$$

One checks (this is an example sheet exercise, and we shall do the general case in Section 9.3) that R_1 is a Riemann surface: away from $w = 0$ the projection $(z, w) \mapsto z$ is a chart, and at the three points $(0, 0), (\pm 1, 0)$ the projection $(z, w) \mapsto w$ is a chart. It contains R as the complement of those three points, so R_1 restores 0 and ± 1. Only ∞ is missing.

To add it, change coordinates so that ∞ becomes finite. Put

$$u = \frac{1}{z}, \qquad v = \frac{z}{w},$$

so that $z = 1/u$ and $w = z/v = 1/(uv)$. Substituting into $w^2 = z^3 - z$:

$$\frac{1}{u^2v^2} = \frac{1}{u^3} - \frac{1}{u} \implies u = v^2(1-u^2).$$

So set

$$R_2 = \{(u,v) \in \mathbb{C}^2 : u = v^2(1-u^2)\},$$

again a Riemann surface by the same argument, and note that $(u,v) = (0,0)$ lies on it – this is our point at infinity. The map

$$\Phi(z,w) = \left(\tfrac{1}{z}, \tfrac{z}{w}\right)$$

is a conformal equivalence from $R_1 \setminus \{(0,0),(\pm 1,0)\}$ onto $R_2 \setminus \{(0,0)\}$, with inverse $(u,v) \mapsto (1/u, 1/(uv))$. Define

$$\hat{R} = R_1 \cup_\Phi R_2.$$

Proposition 8.5

$\hat{R}$ is a compact Riemann surface.

Proof. By Proposition 8.2 we need only Hausdorffness and compactness. Define $\hat{\pi} : \hat{R} \to \mathbb{C}_\infty$ by $\hat{\pi}(i_1(z,w)) = z$ and $\hat{\pi}(i_2(u,v)) = 1/u$; these agree on the overlap by construction of Φ, so $\hat{\pi}$ is a well-defined continuous map, and it is holomorphic in the glued structure.

Hausdorff. Points both in $i_1(R_1)$, or both in $i_2(R_2)$, are separated there. The only points not in $i_2(R_2)$ are $i_1(0,0), i_1(\pm 1, 0)$, and the only point not in $i_1(R_1)$ is $i_2(0,0)$. These are separated by $\hat{\pi}^{-1}(D(0,2))$ and $\hat{\pi}^{-1}(\mathbb{C}_\infty \setminus \overline{D}(0,2))$.

Compact. We show sequential compactness (which suffices, $\hat{R}$ being second countable and metrisable, or directly by a covering argument). Let (p_n) be a sequence in $\hat{R}$. If some subsequence has $|\hat{\pi}(p_n)| \le M$, pass to a further subsequence with $\hat{\pi}(p_n) \to z_0 \in \mathbb{C}$; there are at most two w_0 with $w_0^2 = h(z_0)$, so a further subsequence converges to some $i_1(z_0, w_0)$. Otherwise $|\hat{\pi}(p_n)| \to \infty$, and since $i_2(0,0)$ is the unique point with $\hat{\pi} = \infty$, and $\hat{\pi}$ is a proper map near there, $p_n \to i_2(0,0)$. □

We have obtained a compact Riemann surface $\hat{R}$ carrying a meromorphic function $\hat{\pi}$ (of degree 2, as we shall see) and a meromorphic $\hat{g}$ extending g, with $\hat{g}^2 = \hat{\pi}^3 - \hat{\pi}$. This is the **elliptic curve** $w^2 = z^3 - z$, and in Section 11 we shall discover that it is conformally a complex torus.

§9 Branching and the Degree

The map $\hat{p}_k : \mathbb{C}_\infty \to \mathbb{C}_\infty$, $z \mapsto z^k$, is k-to-one except over 0 and ∞, where the k preimages collide. This is the only local phenomenon there is: we now prove that *every* non-constant holomorphic map looks like $z \mapsto z^n$ in suitable coordinates.

§9.1 The Local Normal Form

Proposition 9.1 (Local Normal Form)

Let $f : R \to S$ be a non-constant holomorphic map of Riemann surfaces and $p \in R$. Then there is an integer $n \geq 1$ and charts (ϕ, U) about p and (ψ, V) about $f(p)$ with $\phi(p) = 0$, $\psi(f(p)) = 0$ and

$$\psi \circ f \circ \phi^{-1}(z) = z^n$$

on $\phi(U)$.

Proof. Start with any charts (ϕ_0, U_0) about p and (ψ, V) about $f(p)$, normalised so that $\phi_0(p) = 0$ and $\psi(f(p)) = 0$, with $f(U_0) \subseteq V$. The local expression $F = \psi \circ f \circ \phi_0^{-1}$ is holomorphic near 0 with $F(0) = 0$, and is not identically zero by Corollary 5.2. So by Proposition 1.3 there is $n \geq 1$ and a holomorphic G near 0 with

$$F(z) = z^n G(z), \qquad G(0) \neq 0.$$

Choose $\delta > 0$ with $D(0, \delta)$ inside the domain of G and $G(D(0, \delta)) \subseteq D(G(0), |G(0)|)$; the latter disc misses 0 and is simply connected, so there is a holomorphic branch ρ of the n-th root on it. Put

$$h(z) = z\,\rho(G(z)), \qquad z \in D(0, \delta),$$

so that $h(z)^n = z^n G(z) = F(z)$. Now $h(0) = 0$ and $h'(0) = \rho(G(0)) \neq 0$, so by the inverse function theorem (Theorem 1.13) h is a biholomorphism from some $D(0, \varepsilon)$ onto an open set.

Set $\phi = h \circ \phi_0$, restricted to $U = \phi_0^{-1}(D(0, \varepsilon))$. This is a chart (a composite of a chart with a biholomorphism) with $\phi(p) = 0$, and it is compatible with the existing structure since h is biholomorphic. Finally

$$\begin{aligned} \psi \circ f \circ \phi^{-1}(z) &= \psi \circ f \circ \phi_0^{-1} \circ h^{-1}(z) \\ &= F(h^{-1}(z)) = h(h^{-1}(z))^n = z^n. \square \end{aligned}$$

Definition 9.2 (Multiplicity)

The integer n of Proposition 9.1 is the **multiplicity** (or **ramification index**, or **local degree**) of f at p, written $\mathrm{mult}_f(p)$.

Lemma 9.3

$\mathrm{mult}_f(p)$ is well defined, i.e. independent of the charts chosen.

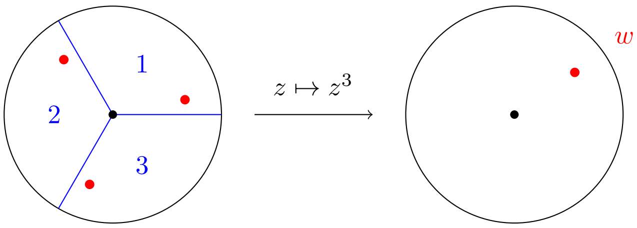
Proof. If z^n and z^m are both local expressions, then they differ by pre-composition and post-composition with biholomorphisms fixing 0. So $\alpha(z)^n = \beta(z^m)$ near 0 with α, β biholomorphic and $\alpha(0) = \beta(0) = 0$. Counting orders of vanishing at 0: α and β have simple zeros (their derivatives are non-zero), so the left side vanishes to order n and the right to order m. Hence $n = m$. □

Equivalently: $\mathrm{mult}_f(p)$ is the order of the zero of $\psi \circ f \circ \phi^{-1} - \psi(f(p))$ at $\phi(p)$, in any charts.

Definition 9.4 (Ramification and Branch Points)

If $\mathrm{mult}_f(p) > 1$ we call p a **ramification point** of f, and $f(p)$ a **branch point**. A point $q \in S$ with $f^{-1}(q)$ a single point p of multiplicity $\deg f$ is **totally ramified**.

The local picture is worth having firmly in mind. Under $z \mapsto z^n$, the disc $|z| < \varepsilon$ is wrapped n times around the disc $|w| < \varepsilon^n$: each of the n sectors of angle $2\pi/n$ is opened out to a full disc, and the n preimages of a point $w \neq 0$ merge into the single preimage 0 of 0.



three preimages one image

Example 9.5

Let $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ be a polynomial $f(z) = a_d z^d + \cdots + a_0$ with $a_d \neq 0$ and $d \geq 1$, extended by $f(\infty) = \infty$. Away from ∞, the local expression of f is f itself and $\mathrm{mult}_f(p)$ is the order of the zero of $f(z) - f(p)$ at p; so the ramification points in $\mathbb{C}$ are exactly the zeros of f'. At ∞, use the chart $w = 1/z$ upstairs and downstairs:

$$\frac{1}{f(1/w)} = \frac{w^d}{a_d + a_{d-1}w + \cdots + a_0 w^d},$$

which vanishes to order d at $w = 0$. So $\mathrm{mult}_f(\infty) = d$, and ∞ is totally ramified whenever $d \geq 2$.

Remark. More generally, if $f : R \to \mathbb{C}$ is holomorphic and (ϕ, U) is any chart about p with $F = f \circ \phi^{-1}$, then writing $F(z) - F(z_0) = (z - z_0)^m G(z)$ with $G(z_0) \neq 0$ and $m = \mathrm{mult}_f(p)$, we get

$$F'(z) = (z - z_0)^{m-1} \left(mG(z) + (z - z_0)G'(z) \right),$$

so $F'(z_0) = 0$ if and only if $m > 1$. The ramification points are the zeros of the derivative in any chart. Since F' is not identically zero (as f is non-constant), its zeros are isolated: *a non-constant holomorphic map has a discrete set of ramification points.* If R is compact, there are only finitely many.

Lemma 9.6 (Multiplicativity)

If $f : R \to S$ and $g : S \to T$ are non-constant holomorphic, then

$$\mathrm{mult}_{g \circ f}(p) = \mathrm{mult}_g(f(p)) \cdot \mathrm{mult}_f(p).$$

Proof. Choose charts in which f is $z \mapsto z^m$ with $m = \mathrm{mult}_f(p)$, and in which g is $w \mapsto w^n$ with $n = \mathrm{mult}_g(f(p))$; we may take the target chart for f to be the source chart for g, since both are just 'some chart about $f(p)$ sending it to 0', and the local normal form allows us to choose the source chart freely at each stage — more carefully, take the normal form charts for g first, then run the proof of Proposition 9.1 for f with that target chart. Then $g \circ f$ has local expression $z \mapsto (z^m)^n = z^{mn}$. □

§9.2 Proper Maps and the Degree

Definition 9.7 (Proper Map)

A continuous map $f : R \to S$ is **proper** if $f^{-1}(K)$ is compact for every compact $K \subseteq S$.

If R is compact then every continuous $f : R \to S$ is proper, since a closed subset of a compact space is compact. That is the case we shall use most often, but properness is the right hypothesis in general: it is what makes fibres finite, and what stops points of a fibre escaping to infinity as we move around in S.

Example 9.8

(i) A non-constant polynomial $f : \mathbb{C} \to \mathbb{C}$ is proper, since $|f(z)| \to \infty$ as $|z| \to \infty$ and so the preimage of a bounded set is bounded and closed.

(ii) $p_k : \mathbb{C}^* \to \mathbb{C}^*$ is proper: $|z^k| \to \infty$ as $|z| \to \infty$ and $|z^k| \to 0$ as $z \to 0$, so preimages of compact subsets of $\mathbb{C}^*$ stay away from both ends.

(iii) *(Non-examples.)* $\exp : \mathbb{C} \to \mathbb{C}^*$ is not proper, since $\exp^{-1}(\{1\}) = 2\pi i\mathbb{Z}$ is an infinite discrete set. The inclusion $\mathbb{D} \hookrightarrow \mathbb{C}$ is not proper either, as $\overline{D}(0, 1)$ has non-compact preimage $\mathbb{D}$. Neither map has a sensible degree: exp is ∞-to-one, and the inclusion is one-to-one over $\mathbb{D}$ and zero-to-one outside it.

The two properties of proper maps we need are the following.

Lemma 9.9

Let $f : R \to S$ be a proper continuous map of Riemann surfaces. Then f is a closed map. If moreover f is holomorphic and non-constant, every fibre $f^{-1}(q)$ is finite.

Proof. For the fibres, $f^{-1}(q)$ is compact, being the preimage of the compact set $\{q\}$, and discrete by Corollary 5.2; a compact discrete space is finite.

For closedness, let $A \subseteq R$ be closed and $q \in \overline{f(A)}$. Pick a chart (ψ, V) about q with $\psi(q) = 0$ and put $N = \psi^{-1}(\overline{D}(0, r))$ for some small $r > 0$ with $\overline{D}(0, r) \subseteq \psi(V)$, a compact neighbourhood of q. Then $f^{-1}(N)$ is compact by properness, so $A \cap f^{-1}(N)$ is compact, and therefore $f\left(A \cap f^{-1}(N)\right)$ is compact and hence closed in the Hausdorff space S. Every neighbourhood of q contained in the interior of N meets $f(A)$, and the points of $f(A)$ it meets lie in $f(A \cap f^{-1}(N))$; so q is in the closure of that set, which is the set itself. Hence $q \in f(A)$, and $f(A)$ is closed. □

Theorem 9.10 (Valency Theorem)

Let $f : R \to S$ be a non-constant proper holomorphic map of Riemann surfaces. Then the function

$$n(q) = \sum_{p \in f^{-1}(q)} \operatorname{mult}_f(p)$$

is constant on S. In particular this applies to any non-constant holomorphic map of compact Riemann surfaces.

Proof. First, each fibre $f^{-1}(q)$ is finite by Lemma 9.9, so n is well defined and $\mathbb{N}$-valued. Since S is connected it suffices to prove n is locally constant.

Fix $q_0 \in S$ and write $f^{-1}(q_0) = \{p_1, \dots, p_k\}$ with $m_i = \operatorname{mult}_f(p_i)$. Choose a chart (ψ, V) about q_0 with $\psi(q_0) = 0$, and by Proposition 9.1 charts (ϕ_i, U_i) about p_i with $\psi f \phi_i^{-1}(z) = z^{m_i}$; shrinking, we may take the U_i pairwise disjoint and contained in $f^{-1}(V)$.

Let $U = \bigcup_i U_i$. Then $R \setminus U$ is closed in R, so $K = f(R \setminus U)$ is closed in S by Lemma 9.9. Note that $q_0 \notin K$, since the whole fibre over q_0 lies in U. Put

$$V' = V \setminus K,$$

an open neighbourhood of q_0, and $U_i' = U_i \cap f^{-1}(V')$. Then

$$f^{-1}(V') \subseteq U, \qquad \text{so} \qquad f^{-1}(V') = \coprod_{i=1}^{k} U_i'.$$

On U_i', in the charts ϕ_i and ψ, f is the map $z \mapsto z^{m_i}$ restricted to an open set; such a map takes each value $w \neq 0$ exactly m_i times (counted without multiplicity, all multiplicities being 1) and the value 0 once with multiplicity m_i. Hence for every $q \in V'$,

$$n(q) = \sum_{i=1}^{k} \sum_{p \in U_i' \cap f^{-1}(q)} \operatorname{mult}_f(p) = \sum_{i=1}^{k} m_i = n(q_0).$$

So n is constant on V'. $\square$

Definition 9.11 (Degree)

The constant value n of Theorem 9.10 is the **degree** of f, written $\deg f$.

Thus a non-constant proper holomorphic map is '$\deg f$-to-one, when points are counted with multiplicity'; in particular this is true of any non-constant holomorphic map of compact Riemann surfaces, which is the situation we shall almost always be in. Note that $p_k : \mathbb{C}^* \to \mathbb{C}^*$ has degree k and a polynomial $\mathbb{C} \to \mathbb{C}$ has degree equal to its degree as a polynomial, so the notion is consistent with the classical one even off compact surfaces.

Corollary 9.12

A holomorphic map $f : R \to S$ of compact Riemann surfaces with $\deg f = 1$ is a conformal equivalence. In particular a holomorphic bijection of compact Riemann surfaces is a conformal equivalence.

Proof. If $\deg f = 1$ then every fibre is a single point of multiplicity 1, so f is a bijection with no ramification points. By Proposition 9.1 f is locally biholomorphic, so f^{-1} is holomorphic. $\square$

Example 9.13

If f is a polynomial of degree d, regarded as $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$, then $\deg f = d$: compute $n(\infty)$, which by Example 9.5 is $\mathrm{mult}_f(\infty) = d$ since ∞ is the only preimage of ∞.

Corollary 9.14 (Fundamental Theorem of Algebra)

A polynomial $f \in \mathbb{C}[z]$ of degree $d \geq 1$ has exactly d roots in $\mathbb{C}$, counted with multiplicity.

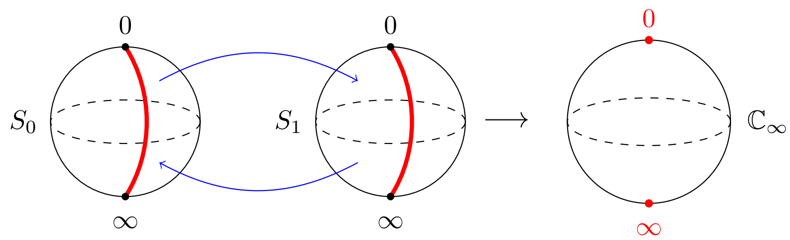
Proof. Extend to $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$, of degree d. Then $n(0) = d$, and $f^{-1}(0) \subseteq \mathbb{C}$ since $f(\infty) = \infty$. The multiplicity of f at a root is the order of the root. $\square$

Example 9.15

A rational function $f = p/q$ in lowest terms with $\deg p = m$, $\deg q = n$ has $\deg f = \max(m, n)$ as a map $\mathbb{C}_\infty \to \mathbb{C}_\infty$. Indeed if $m > n$ then ∞ is the only preimage of ∞ with multiplicity $m - n$... rather than pursue this, compute $n(w)$ for generic w: the equation $p(z) - wq(z) = 0$ has $\max(m, n)$ roots for all but finitely many w, since the leading coefficient of $p - wq$ vanishes for at most one w.

Example 9.16 (The Two-Sheeted Cover of the Sphere)

Consider $\hat{p}_2 : \mathbb{C}_\infty \to \mathbb{C}_\infty$, $z \mapsto z^2$. It has degree 2, and its branch points are 0 and ∞, each totally ramified. Topologically: take two copies of the sphere, slit each along an arc from 0 to ∞, and glue crosswise. The result is again a sphere.



Gluing the upper side of the slit in S_0 to the lower side of the slit in S_1, and vice versa, returns a sphere; the two red points are the branch points 0 and ∞, and over every other point there are two preimages. That the total space is again a sphere and not, say, a torus is a genuine constraint, and Riemann–Hurwitz is the tool that makes such constraints computable.

§9.3 Riemann Surfaces of Algebraic Functions

The examples $w^2 = z^3 - z$ and $w^k = z$ are instances of a general construction. Let

$$P(z, w) = w^d + c_{d-1}(z)w^{d-1} + \cdots + c_0(z)$$

be a polynomial in w whose coefficients are polynomials in z, and suppose P is irreducible in $\mathbb{C}[z, w]$. The **algebraic function** $w(z)$ defined by $P(z, w) = 0$ is the classical object; the modern one is the surface

$$X' = \{(z, w) \in \mathbb{C}^2 : P(z, w) = 0\}.$$

Proposition 9.17

Suppose X' is **smooth**, that is, at every point of X' at least one of $\partial P/\partial z$, $\partial P/\partial w$ is non-zero. Then X' is a Riemann surface, with charts given by the coordinate projections, and $\pi_z : X' \to \mathbb{C}$, $(z, w) \mapsto z$, is holomorphic.

Proof sketch. X' is Hausdorff and second countable as a subspace of $\mathbb{C}^2$. Near a point (z_0, w_0) with $\partial P/\partial w \neq 0$, the holomorphic implicit function theorem provides a holomorphic η near z_0 with $\eta(z_0) = w_0$ and $P(z, \eta(z)) = 0$, and the graph of η is a neighbourhood of (z_0, w_0) in X'; so π_z is a homeomorphism there and serves as a chart. Symmetrically, where $\partial P/\partial z \neq 0$, π_w is a chart. Smoothness says these cover X'. The transition functions are η and its inverse, which are holomorphic. Connectedness is the one genuinely non-trivial point, and it follows from irreducibility of P; we verify it by hand in the examples we need. □

Away from the finitely many z for which $P(z, \cdot)$ has a repeated root – the **discriminant locus** – the map π_z is a d-sheeted covering map, since the d roots w vary holomorphically and distinctly. Over the discriminant locus, roots collide and π_z ramifies. Compactifying (by gluing in charts at ∞, as in Section 8.3) gives a compact Riemann surface X with a degree-d meromorphic function $\hat{\pi}_z : X \to \mathbb{C}_\infty$.

Example 9.18 (Fermat Curves)

Let $d \geq 1$ and

$$F_d' = \{(x, y) \in \mathbb{C}^2 : x^d + y^d = 1\}.$$

Here $P = y^d + x^d - 1$, with $\partial P/\partial y = dy^{d-1}$ and $\partial P/\partial x = dx^{d-1}$, which vanish simultaneously only at $(0, 0)$, not a point of F_d'. So F_d' is smooth.

Concretely, π_x has local inverse $x \mapsto (x, \sqrt[d]{1 - x^d})$, which exists and is holomorphic near any x_0 with $x_0^d \neq 1$, i.e. away from $y_0 = 0$; and π_y provides charts near the points where $y = 0$, since there $x \neq 0$. These cover F_d'.

For connectedness: let

$$D = \mathbb{C} \setminus \bigcup_{i=0}^{d-1} \{t\zeta_d^i : t \geq 1\}, \qquad \zeta_d = e^{2\pi i/d},$$

the plane with d radial slits removed. This is simply connected and $1 - x^d$ does not vanish on it, so there are d holomorphic branches of $(1 - x^d)^{1/d}$ on D, each extending continuously to the slit endpoints ζ_d^i. Given $(x_0, y_0) \in F_d'$ with $x_0 \in D$, choose the branch y with $y(x_0) = y_0$ and a path γ in D from x_0 to 1; then $t \mapsto (\gamma(t), y(\gamma(t)))$ joins (x_0, y_0) to $(1, 0)$. Points with $x_0 \notin D$ can first be joined to nearby points with $x_0 \in D$, as D is dense. So F'_d is path-connected, hence a Riemann surface.

The compactification F_d is obtained by gluing on a second chart at infinity (an example sheet exercise), and $\hat{\pi}_x : F_d \to \mathbb{C}_\infty$ has degree d. Its ramification points are the d points $(\zeta_d^i, 0)$, each of multiplicity d (there $1 - x^d$ has a simple zero, so $y = (1 - x^d)^{1/d}$ is a d-th root of a simple zero), and $\hat{\pi}_x^{-1}(\infty)$ consists of d distinct unramified points. We compute the genus of F_d in Example 10.12.

§10 Euler Characteristic and Riemann–Hurwitz

Everything so far has been local or analytic. We now bring in topology, and the payoff is immediate: the degree and ramification of a holomorphic map determine the topology of its source from that of its target.

§10.1 Triangulations and the Euler Characteristic

Definition 10.1 (Triangulation)

Let S be a compact Riemann surface. A **topological triangle** in S is a continuous embedding $\Delta \hookrightarrow S$ of a closed non-degenerate triangle $\Delta \subseteq \mathbb{R}^2$. A **triangulation** of S is a finite collection $\{\Delta_i\}$ of topological triangles such that

(i) $\bigcup_i \Delta_i = S$;

(ii) for $i \neq j$, $\Delta_i \cap \Delta_j$ is empty, a common vertex, or a common edge;

(iii) each edge lies in exactly two triangles.

Definition 10.2 (Euler Characteristic)

The **Euler characteristic** of a triangulation is $\chi = V - E + F$, where V, E, F count vertices, edges and faces.

Lemma 10.3

Every compact Riemann surface admits a triangulation, and χ is independent of the triangulation chosen.

We take this on trust; the first part is Radó's theorem and the second is the topological invariance of simplicial homology, proved in II Algebraic Topology (where $\chi = \sum_i (-1)^i \operatorname{rank} H_i$). We write $\chi(S)$ for the common value.

Example 10.4

$\chi(\mathbb{C}_\infty) = 2$. Triangulate the sphere as the boundary of a tetrahedron: $V - E + F = 4 - 6 + 4 = 2$.

Example 10.5

$\chi(\mathbb{C}/\Lambda) = 0$. Represent the torus as a square with opposite sides identified and cut it into two triangles by a diagonal. After identification there is $V = 1$ vertex, $E = 3$ edges (two sides and the diagonal), and $F = 2$ faces, giving $1 - 3 + 2 = 0$. (Strictly this is not a triangulation in the sense above, since the two triangles share more than one edge; subdividing repairs this without changing χ.)

We quote from II Algebraic Topology the classification of surfaces: every compact connected orientable surface is homeomorphic to Σ_g, the g-holed torus, for a unique $g \geq 0$, and

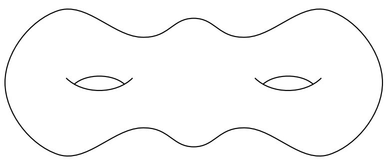
$$\chi(\Sigma_g) = 2 - 2g.$$

A Riemann surface is orientable, since holomorphic transition functions have positive Jacobian determinant $|\phi'|^2 > 0$. So a compact Riemann surface is homeomorphic to some Σ_g, and χ determines it up to homeomorphism.

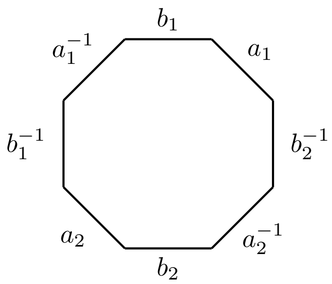
Definition 10.6 (Genus)

The **genus** of a compact Riemann surface S is $g(S) = (2 - \chi(S))/2$.

The standard picture of Σ_g is a sphere with g handles; equivalently it is a $4g$-gon with its edges identified in the pattern $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$. For $g = 2$:



Σ_2



the 8-gon

From the octagon picture one reads off $\chi(\Sigma_2)$ directly: all 8 vertices are identified to one, the 8 edges become 4, and there is 1 face, giving $1 - 4 + 1 = -2 = 2 - 2 \cdot 2$.

§10.2 The Riemann–Hurwitz Formula

Theorem 10.7 (Riemann–Hurwitz)

Let $f : R \to S$ be a non-constant holomorphic map of compact Riemann surfaces of degree n. Then

$$\chi(R) = n\,\chi(S) - \sum_{p \in R} (\mathrm{mult}_f(p) - 1)\,.$$

Equivalently, in terms of genera,

$$2g(R) - 2 = n\,(2g(S) - 2) + \sum_{p \in R} (\mathrm{mult}_f(p) - 1)\,.$$

The sum is finite: as we observed above, the ramification points of f are isolated, and R is compact, so there are finitely many.

Proof. As in the proof of the valency theorem, every $q \in S$ has an open neighbourhood V_q which is **well covered**: $f^{-1}(V_q)$ is a disjoint union of open sets on each of which f is, in suitable charts, a power map $z \mapsto z^m$, and V_q contains no branch point other than possibly q. By compactness of S, finitely many $V_{q_1}, \ldots, V_{q_r}$ cover S; in particular there are finitely many branch points.

Choose a triangulation of S and subdivide it repeatedly (barycentrically, say) until

(i) every branch point of f is a vertex, and

(ii) every triangle is contained in some V_{q_i}.

Both are achievable: (i) by first taking a triangulation with the finitely many branch points among the vertices, and (ii) since subdivision makes the triangles arbitrarily small and the V_{q_i} have a Lebesgue number.

Now pull back. If $\Delta \subseteq V_{q_i}$ is a (closed) triangle then $f^{-1}(\Delta)$ is a disjoint union of triangles, each mapped homeomorphically onto Δ by f; the same for edges. Indeed Δ is simply connected and, since its interior contains no branch point, f restricted over the interior is a covering map of degree n; and the branch points on $\partial\Delta$ are vertices, where the local picture $z \mapsto z^m$ still sends the m triangle-corners upstairs homeomorphically onto the corner downstairs. So the preimages of the triangles of S form a triangulation of R, with

$$F_R = nF_S, \qquad E_R = nE_S,$$

since each face and each edge downstairs has exactly n preimages.

Vertices are where the count fails. For $q \in S$ a vertex, the valency theorem gives

$$\begin{aligned} \left|f^{-1}(q)\right| = \sum_{p \in f^{-1}(q)} 1 &= \sum_{p \in f^{-1}(q)} \operatorname{mult}_f(p) - \sum_{p \in f^{-1}(q)} \left(\operatorname{mult}_f(p) - 1\right) \\ &= n - \sum_{p \in f^{-1}(q)} \left(\operatorname{mult}_f(p) - 1\right). \end{aligned}$$

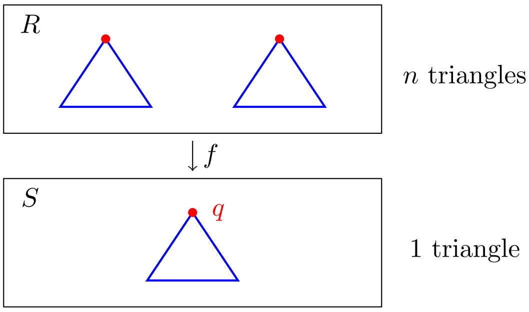
Summing over the vertices q of S, and noting that every ramification point of f lies over a branch point, which is a vertex,

$$V_R = nV_S - \sum_{p \in R} \left(\operatorname{mult}_f(p) - 1\right).$$

Therefore

$$\chi(R) = V_R - E_R + F_R = n(V_S - E_S + F_S) - \sum_{p \in R} (\operatorname{mult}_f(p) - 1),$$

which is the first formula. The second follows by substituting $\chi = 2 - 2g$. $\square$



Each triangle and each edge downstairs has exactly n preimages; only the vertices lying under ramification points have fewer than n preimages, and the deficiency is exactly $\sum(\text{mult}_f(p) - 1)$.

Remark. The correction term $\sum_p(\text{mult}_f(p) - 1)$ is always even, being $\chi(R) - n\chi(S) = (2 - 2g(R)) - n(2 - 2g(S))$. This trivial-looking observation is surprisingly useful: it often pins down the ramification at a point we have not analysed.

§10.3 First Applications

Corollary 10.8

Suppose $f : R \to S$ is an unramified non-constant holomorphic map of compact Riemann surfaces (equivalently, a covering map) of degree n. Then $g(R) - 1 = n(g(S) - 1)$, and

(i) if $g(S) = 0$ then $n = 1$ and $R \cong S \cong \mathbb{C}_\infty$;

(ii) if $g(S) = 1$ then $g(R) = 1$, and n is unrestricted;

(iii) if $g(S) \geq 2$ then either $n = 1$ and f is a conformal equivalence, or $n \geq 2$ and $g(R) > g(S)$.

Proof. Riemann–Hurwitz with no correction term gives $2g(R) - 2 = n(2g(S) - 2)$. If $g(S) = 0$ then $g(R) - 1 = -n < 0$, forcing $g(R) = 0$ and $n = 1$; then f is a conformal equivalence by Corollary 9.12, and a genus 0 compact surface is $\mathbb{C}_\infty$ (which we prove in Corollary 13.8). If $g(S) = 1$ then $g(R) = 1$ for any n. If $g(S) \geq 2$ then $g(R) - 1 = n(g(S) - 1) \geq n$, so $n \geq 2$ gives $g(R) \geq 2g(S) - 1 > g(S)$. □

Example 10.9

That case (ii) is genuinely unrestricted: take $\Lambda_n = \langle n, i\rangle$ for $n \in \mathbb{Z}_{>0}$. Then $\Lambda_n \leq \Lambda_1 = \langle 1, i\rangle$ with index n, so the identity on $\mathbb{C}$ descends to an unramified degree-n map $\mathbb{C}/\Lambda_n \to \mathbb{C}/\Lambda_1$ of tori.

Example 10.10 (The Cubic Curve)

Let $\hat{R}$ be the compactification of the surface of $w^2 = z^3 - z$ from Section 8.3, with $\hat{\pi} : \hat{R} \to \mathbb{C}_\infty$. Since a generic z has exactly two preimages $(z, \pm\sqrt{h(z)})$, $\deg \hat{\pi} = 2$. The branch points are $0, \pm 1$ (where h has a simple zero, so the two sheets come together) and ∞; each is totally ramified, with a single preimage of multiplicity 2.

So
$$2g(\hat{R}) - 2 = 2(0-2) + 4 \cdot (2-1) = -4 + 4 = 0 \implies g(\hat{R}) = 1,$$
confirming that $\hat{R}$ is a torus.

Remark. Suppose we had not analysed the behaviour at ∞. Writing
$$C = \sum_{p \in \hat{\pi}^{-1}(\infty)} \left(\operatorname{mult}_{\hat{\pi}}(p) - 1\right),$$
the correction term is $3 + C$, which must be even by the parity remark above; so C is odd, and since $\deg \hat{\pi} = 2$ we must have $C = 1$, i.e. ∞ is a branch point. The parity argument has done the local analysis for us. This works whenever $\deg f = 2$.

Example 10.11 (Hyperelliptic Curves)
More generally, let $h(z)$ be a polynomial of degree d with d distinct roots, and let X be the compactification of $\{w^2 = h(z)\}$, with $\hat{\pi} : X \to \mathbb{C}_\infty$ of degree 2. The branch points in $\mathbb{C}$ are the d roots of h. What happens at ∞? Setting $u = 1/z$, we get $w^2 = u^{-d}(1 + O(u))$, so $w \sim u^{-d/2}$: if d is even there are two points over ∞, unramified; if d is odd there is one, ramified. Either way the total correction is d or $d+1$, i.e. $2\lceil d/2 \rceil$, and
$$2g(X) - 2 = 2(-2) + 2\left\lceil \tfrac{d}{2} \right\rceil \implies g(X) = \left\lceil \tfrac{d}{2} \right\rceil - 1 = \left\lfloor \frac{d-1}{2} \right\rfloor.$$
So $d = 3, 4$ give genus 1; $d = 5, 6$ give genus 2; and so on. These are the **hyperelliptic** curves.

Example 10.12 (Fermat Curves and the Degree–Genus Formula)
Let F_d be the compactified Fermat curve of Example 9.18, with $\hat{\pi}_x : F_d \to \mathbb{C}_\infty$ of degree d. There are d ramification points $(\zeta_d^i, 0)$, each of multiplicity d, and no others. So
$$2g(F_d) - 2 = d(0-2) + d(d-1) \implies g(F_d) = \frac{(d-1)(d-2)}{2}.$$

This is an instance of the **degree–genus formula**: a smooth projective plane curve of degree d has genus $(d-1)(d-2)/2$. For $d = 1, 2$ we get $g = 0$ (lines and conics are spheres); for $d = 3$, $g = 1$ (cubics are tori, as we found for $w^2 = z^3 - z$ after homogenising); for $d = 4$, $g = 3$; and g grows quadratically. In particular there exist compact Riemann surfaces of arbitrarily large genus – something not at all obvious from the definitions.

Note that not every genus arises from a plane curve: $(d-1)(d-2)/2$ takes the values $0, 0, 1, 3, 6, 10, \dots$, so no smooth plane curve has genus 2. (Genus 2 surfaces exist all the same – they are hyperelliptic, by the previous example with $d = 5$.)

§11 Elliptic Functions and Complex Tori

We now describe all the meromorphic functions on the two compact surfaces we understand best: the sphere, and the torus. The sphere is easy and classical. The torus is the heart of the course, and the answer is remarkable – every complex torus is a plane cubic curve, and every elliptic function is a rational expression in one particular function and its derivative.

§11.1 Meromorphic Functions on the Sphere

Proposition 11.1

Every meromorphic function $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ is rational:

$$f(z) = c\,\frac{(z-a_1)\cdots(z-a_m)}{(z-b_1)\cdots(z-b_n)}$$

for some $c \in \mathbb{C}^*$ and $a_i, b_j \in \mathbb{C}$ with $a_i \neq b_j$.

Proof. We may assume f is non-constant. Replacing f by $1/f$ if necessary (which does not affect the conclusion), assume $f(\infty) \neq \infty$. The poles of f form a discrete closed subset of the compact space $\mathbb{C}_\infty$, hence a finite set $\{b_1, \dots, b_{n'}\}$, and none is ∞. About each b_j write the Laurent expansion

$$f(z) = \sum_{l \geq -k_j} c_{j,l}(z-b_j)^l, \qquad c_{j,-k_j} \neq 0,$$

and let $Q_j(z) = \sum_{l=-k_j}^{-1} c_{j,l}(z-b_j)^l$ be the principal part. Then

$$g = f - \sum_{j=1}^{n'} Q_j$$

is holomorphic on all of $\mathbb{C}$ – the principal parts have cancelled every pole – and is bounded near ∞, since f is (having a finite value there) and each $Q_j \to 0$. By Liouville, g is constant. So f is a constant plus a sum of principal parts, which is a rational function. $\square$

Remark. If f is written as above in lowest terms, its degree as a holomorphic map is $\max(m, n)$: this counts the preimages of a generic point. The zeros a_i and poles b_j may be prescribed arbitrarily, subject only to being finite in number – there is no constraint on the sphere. On a torus we shall find that there is one.

§11.2 Periodic Meromorphic Functions

Definition 11.2 (Period)

Let $f : \mathbb{C} \to \mathbb{C}_\infty$ be meromorphic. A **period** of f is $\omega \in \mathbb{C}$ with $f(z+\omega) = f(z)$ for all z.

The set Ω of periods is a subgroup of $(\mathbb{C}, +)$, and it is closed: if $\omega_n \to \omega$ with $\omega_n \in \Omega$ then $f(z+\omega) = f(z)$ for all z not a pole, by continuity, and hence everywhere.

Lemma 11.3

Let f be a non-constant meromorphic function on $\mathbb{C}$ with period group Ω. Then exactly one of the following holds:

(i) $\Omega = \{0\}$;

(ii) $\Omega = \langle \omega \rangle \cong \mathbb{Z}$ for some $\omega \neq 0$;

(iii) $\Omega = \langle \omega_1, \omega_2 \rangle \cong \mathbb{Z}^2$ is a lattice, with $\omega_1/\omega_2 \notin \mathbb{R}$.

(If f is constant then $\Omega = \mathbb{C}$.)

Proof sketch. Ω is a closed subgroup of $\mathbb{C} \cong \mathbb{R}^2$, and the classification of such subgroups gives $\{0\}$, a discrete rank-1 or rank-2 subgroup, a line, a line plus a discrete complement, or all of $\mathbb{C}$. If Ω contained a line $\mathbb{R}\omega$ then the zeros of $f - f(z_0)$ would be non-discrete for suitable z_0, contradicting the identity theorem. The remaining possibilities are (i)–(iii). This is on the example sheets. □

Definition 11.4

A non-constant meromorphic f on $\mathbb{C}$ is **simply periodic** in case (ii), and **doubly periodic** or **elliptic** in case (iii).

The reason to care is the correspondence with functions on quotients.

Proposition 11.5

Let f be meromorphic on $\mathbb{C}$ and suppose $\Lambda \subseteq \Omega$ where $\Lambda \leq \mathbb{C}$ is a lattice (respectively, $\langle \omega \rangle \subseteq \Omega$). Then there is a unique meromorphic function $\bar{f}$ on $\mathbb{C}/\Lambda$ (respectively on $\mathbb{C}^*$) with

$$f = \bar{f} \circ \pi, \qquad \pi : \mathbb{C} \to \mathbb{C}/\Lambda \quad (\text{resp. } z \mapsto e^{2\pi i z/\omega}).$$

Proof. Uniqueness is clear as π is surjective. For existence set $\bar{f}(\pi(z)) = f(z)$, which is well defined by periodicity. It is holomorphic as a map to $\mathbb{C}_\infty$ because π is a covering map with local holomorphic inverses: near any point, $\bar{f} = f \circ (\pi|_{\tilde{U}})^{-1}$. □

Thus simply periodic functions are the same thing as meromorphic functions on $\mathbb{C}^* \cong \mathbb{C}/\langle \omega \rangle$, and elliptic functions with $\Lambda \subseteq \Omega$ are the same thing as meromorphic functions on the complex torus $\mathbb{C}/\Lambda$. Everything we know about compact surfaces now applies.

Corollary 11.6

A holomorphic doubly periodic function on $\mathbb{C}$ is constant.

Proof. It descends to a holomorphic function on the compact surface $\mathbb{C}/\Lambda$, which is constant by Corollary 5.5. □

Corollary 11.7

A non-constant elliptic function, viewed as $\bar{f} : \mathbb{C}/\Lambda \to \mathbb{C}_\infty$, has $\deg \bar{f} \geq 2$.

Proof. If $\deg \bar{f} = 1$ then $\bar{f}$ is a conformal equivalence $\mathbb{C}/\Lambda \to \mathbb{C}_\infty$ by Corollary 9.12, contradicting $\chi(\mathbb{C}/\Lambda) = 0 \neq 2 = \chi(\mathbb{C}_\infty)$. (Equivalently: Riemann–Hurwitz would give $0 = 1 \cdot (-2) + 0$.) $\square$

It is worth also seeing the classical proof, which is where the theory came from.

Second proof of Corollary 11.7. Choose z_0 so that no zero or pole of f lies on the boundary of the period parallelogram $P = z_0 + \{s\omega_1 + t\omega_2 : s, t \in [0,1]\}$; this is possible as there are finitely many such points modulo Λ. The residue theorem gives

$$\sum_{z \text{ pole in } P^\circ} \operatorname{res}_z(f) = \frac{1}{2\pi i} \oint_{\partial P} f(z)\, \mathrm{d}z = 0,$$

because the contributions of opposite sides of P cancel: f takes the same values on them and they are traversed in opposite directions. If f had a single simple pole in P, its residue would be non-zero, contradiction. So f has either no poles (hence is constant) or at least two poles counted with multiplicity. $\square$

§11.3 The Weierstrass $\wp$-Function

We need a non-constant elliptic function to exist at all. By the above the simplest possible has a single double pole per period parallelogram, and this pins down the definition.

Definition 11.8 (Weierstrass $\wp$-function)
Let $\Lambda \subseteq \mathbb{C}$ be a lattice. The **Weierstrass $\wp$-function** is

$$\wp(z) = \wp_\Lambda(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right).$$

The subtracted terms $1/\omega^2$ are there purely to make the series converge; without them it diverges. To prove convergence we need a lemma on lattice sums.

Lemma 11.9
Let $\Lambda = \langle \omega_1, \omega_2 \rangle$ be a lattice and $t \in \mathbb{R}$. Then

$$\sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{|\omega|^t}$$

converges if and only if $t > 2$.

Proof. Consider $Q = \{(t_1, t_2) \in \mathbb{R}^2 : |t_1| + |t_2| = 1\}$, a compact set, and the continuous function $(t_1, t_2) \mapsto |t_1\omega_1 + t_2\omega_2|$ on it. It attains a maximum M and a minimum m, and $m > 0$ since ω_1, ω_2 are linearly independent over $\mathbb{R}$ so the expression vanishes only at the origin, which is not in Q. Given $(k, l) \in \mathbb{Z}^2 \setminus \{0\}$, apply this with $t_1 = k/(|k| + |l|)$, $t_2 = l/(|k| + |l|)$ to get

$$m(|k| + |l|) \leq |k\omega_1 + l\omega_2| \leq M(|k| + |l|).$$

So the sum in question is bounded above and below by positive multiples of

$$\sum_{(k,l)\neq(0,0)} \frac{1}{(|k|+|l|)^t}.$$

For each $n \geq 1$ there are exactly $4n$ pairs (k,l) with $|k|+|l| = n$, so this equals

$$\sum_{n\geq 1} \frac{4n}{n^t} = 4\sum_{n\geq 1} \frac{1}{n^{t-1}},$$

which converges if and only if $t-1 > 1$. □

Notice that the same estimate shows every bounded region contains only finitely many lattice points, which is what we assumed in Proposition 4.26.

Theorem 11.10

$\wp_\Lambda$ is a well-defined elliptic function whose period group is exactly Λ. It is even, its poles are precisely the points of Λ, each a double pole, and $\deg \wp_\Lambda = 2$ as a map $\mathbb{C}/\Lambda \to \mathbb{C}_\infty$.

Proof. Convergence. Fix $R > 0$ and work on $|z| \leq R$. For $\omega \neq 0$,

$$\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} = \frac{\omega^2 - (z-\omega)^2}{\omega^2(z-\omega)^2} = \frac{z(2\omega - z)}{\omega^2(z-\omega)^2} = \frac{z}{\omega^2}\left(\frac{2}{z-\omega} + \frac{-z}{(z-\omega)^2}\right)\cdot(-1),$$

so

$$\left|\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right| \leq \frac{|z|}{|\omega|^2}\left(\frac{2}{|z-\omega|} + \frac{|z|}{|z-\omega|^2}\right).$$

For all but finitely many ω we have $|\omega| \geq 2R$, whence $|z-\omega| \geq |\omega| - R \geq |\omega|/2 \geq R$, and the bound becomes

$$\frac{R}{|\omega|^2}\left(\frac{4}{|\omega|} + \frac{R}{(|\omega|/2)\cdot R}\right) = \frac{R}{|\omega|^2}\cdot\frac{6}{|\omega|} = \frac{6R}{|\omega|^3}.$$

By Lemma 11.9 with $t = 3$, the series converges absolutely and uniformly on $|z| \leq R$ after discarding finitely many terms. So $\wp$ is a well-defined meromorphic function on $\mathbb{C}$, holomorphic away from Λ, with a double pole at each lattice point (the discarded terms contribute those).

Evenness. Replacing z by $-z$ and ω by $-\omega$ permutes the terms of the (absolutely convergent) sum, so $\wp(-z) = \wp(z)$.

Periodicity. Differentiating term by term (legitimate by local uniform convergence),

$$\wp'(z) = -2\sum_{\omega\in\Lambda} \frac{1}{(z-\omega)^3},$$

which is manifestly Λ-periodic, since translating z by $\omega_0 \in \Lambda$ just permutes the summands. Fix $\omega_0 \in \Lambda$ and put $F(z) = \wp(z+\omega_0) - \wp(z)$. Then $F' = 0$, so F is constant, say $F \equiv c$. Now take $z = -\omega_0/2$, which is not a pole if $\omega_0/2 \notin \Lambda$ (and if it is, replace ω_0 by a generator and argue for generators, which suffices):

$$c = \wp(\omega_0/2) - \wp(-\omega_0/2) = 0$$

by evenness. So ω_0 is a period, and $\Lambda \subseteq \Omega$.

Conversely the poles of $\wp$ are exactly Λ; a period must permute the poles, so $\Omega \subseteq \Lambda$ (as 0 is a pole, any $\omega \in \Omega$ satisfies $-\omega \in \Lambda$). Hence $\Omega = \Lambda$.

Degree. On $\mathbb{C}/\Lambda$, $\wp$ has a single pole, at 0, of order 2. So $\deg \wp = n(\infty) = 2$. □

Remark. The properties

(i) $\wp$ is meromorphic with period lattice Λ,

(ii) the poles of $\wp$ are exactly Λ, and

(iii) $\wp(z) - z^{-2} \to 0$ as $z \to 0$

characterise $\wp$ uniquely: if $\tilde{\wp}$ also has them then $\wp - \tilde{\wp}$ is holomorphic and elliptic, hence constant, and the constant is 0 by (iii).

§11.4 Ramification of $\wp$

Since $\deg \wp = 2$, Riemann–Hurwitz gives

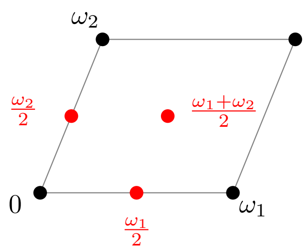
$$0 = 2 \cdot (-2) + \sum_p (\mathrm{mult}_\wp(p) - 1) \implies \sum_p (\mathrm{mult}_\wp(p) - 1) = 4,$$

so $\wp$ has exactly four ramification points, each of multiplicity 2 (multiplicity cannot exceed the degree). Let us find them.

One is $p = 0$: the pole is double, so $\mathrm{mult}_\wp(0) = 2$. The others are the zeros of $\wp'$, since ramification points are the zeros of the derivative in a chart. Now $\wp'$ is odd (differentiate $\wp(-z) = \wp(z)$), elliptic with period lattice Λ, and has a triple pole at each lattice point, so $\deg \wp' = 3$ on $\mathbb{C}/\Lambda$. For any $\omega \in \Lambda$,

$$\wp'(\omega/2) = \wp'(\omega/2 - \omega) = \wp'(-\omega/2) = -\wp'(\omega/2),$$

using periodicity and oddness; hence $\wp'(\omega/2) = 0$ whenever $\omega/2 \notin \Lambda$. Taking $\omega = \omega_1, \omega_2, \omega_1+\omega_2$ gives three distinct zeros in $\mathbb{C}/\Lambda$, namely the three non-zero **half-periods**. Since $\deg \wp' = 3$ there are no others, and all three are simple.



Proposition 11.11

The four ramification points of $\wp : \mathbb{C}/\Lambda \to \mathbb{C}_\infty$ are $0, \omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$, each of multiplicity 2, and their images

$$\infty, \quad e_1 = \wp(\omega_1/2), \quad e_2 = \wp(\omega_2/2), \quad e_3 = \wp\left(\tfrac{\omega_1+\omega_2}{2}\right)$$

are four distinct points of $\mathbb{C}_\infty$.

Proof. We have found the four ramification points and know each has multiplicity 2. They must have distinct images: if two of them mapped to the same q, then $n(q)$ would be at least $4 > 2 = \deg \wp$. $\square$

So $\wp$ realises $\mathbb{C}/\Lambda$ as a two-sheeted cover of the sphere branched over four points – exactly the description of the surface of $w^2 = z^3 - z$ found in Example 10.10. This is not a coincidence.

§11.5 The Differential Equation

Proposition 11.12

There are constants $g_2, g_3 \in \mathbb{C}$, depending only on Λ, such that

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

Proof. Near $z = 0$ we expand. Since $\wp$ is even and $\wp(z) - z^{-2}$ vanishes at 0, its Laurent series has the form

$$\wp(z) = \frac{1}{z^2} + az^2 + bz^4 + O(z^6)$$

(no odd powers, and no constant term). Cubing,

$$\wp(z)^3 = \frac{1}{z^6} + \frac{3a}{z^2} + 3b + O(z^2),$$

while differentiating and squaring,

$$\wp'(z) = \frac{-2}{z^3} + 2az + 4bz^3 + O(z^5), \qquad \wp'(z)^2 = \frac{4}{z^6} - \frac{8a}{z^2} - 16b + O(z^2).$$

Hence

$$\wp'(z)^2 - 4\wp(z)^3 = -\frac{20a}{z^2} - 28b + O(z^2).$$

Set $g_2 = 20a$. Then

$$H := (\wp')^2 - 4\wp^3 + g_2\wp$$

is elliptic, and near 0 it equals $-28b + O(z^2) + g_2 \cdot O(z^2)$, so it is holomorphic at 0. Its only possible poles were at Λ, so H is a holomorphic elliptic function, hence constant; call the constant $-g_3$. Rearranging gives the result. $\square$

Remark. The constants are computable: expanding $\wp$ gives $g_2 = 60\sum_{\omega\neq 0}\omega^{-4}$ and $g_3 = 140\sum_{\omega\neq 0}\omega^{-6}$, which converge by Lemma 11.9. We shall not need this.

Corollary 11.13

e_1, e_2, e_3 are the three roots of $4X^3 - g_2X - g_3$; they are distinct, and $e_1+e_2+e_3 = 0$. Moreover
$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. At $z = \omega_i/2$ we have $\wp'(z) = 0$ and $\wp(z) = e_i$, so $0 = 4e_i^3 - g_2e_i - g_3$. The e_i are distinct by Proposition 11.11, so they are all three roots; the sum of the roots is $-(\text{coefficient of } X^2)/4 = 0$. The factorisation follows. □

§11.6 The Torus as a Cubic Curve

Theorem 11.14

Let $\Lambda \subseteq \mathbb{C}$ be a lattice with invariants g_2, g_3 as above, and let

$$X' = \{(x, y) \in \mathbb{C}^2 : y^2 = 4x^3 - g_2x - g_3\},$$

with $X = X' \cup \{\infty\}$ its one-point compactification (a compact Riemann surface, as in Section 8.3). Then
$$\mathbb{C}/\Lambda \cong X$$
as Riemann surfaces, via $z + \Lambda \mapsto (\wp(z), \wp'(z))$ and $0 \mapsto \infty$.

Proof. Since $4X^3 - g_2X - g_3 = 4(X - e_1)(X - e_2)(X - e_3)$ has distinct roots, X' is smooth and is a Riemann surface with coordinate projections as charts (Proposition 9.17), and X is its compactification.

Define $F : \mathbb{C} \setminus \Lambda \to X'$ by $F(z) = (\wp(z), \wp'(z))$. Its image lies in X' by Proposition 11.12. It is holomorphic: composed with the chart π_x it is $\wp$, and composed with π_y it is $\wp'$, both holomorphic. Extending by $F(\omega) = \infty$ for $\omega \in \Lambda$ gives a holomorphic map $\mathbb{C} \to X$ (the point ∞ of X is exactly where both coordinates blow up, and the local computation there is as in Section 8.3). Since F is Λ-periodic it descends, by Proposition 11.5, to a holomorphic

$$\Phi : \mathbb{C}/\Lambda \to X.$$

Φ is non-constant, and $\mathbb{C}/\Lambda$ is compact, so Φ is surjective by Corollary 5.4. It remains to prove injectivity, for then $\deg \Phi = 1$ and Corollary 9.12 finishes the proof.

Work in the centred period parallelogram P with vertices $\pm(\omega_1 \pm \omega_2)/2$, whose interior P° maps injectively to $\mathbb{C}/\Lambda$ and contains exactly one point of each class except for those on ∂P. Suppose $\Phi(z) = \Phi(z')$ for $z, z' \in P^\circ$. Then $\wp(z) = \wp(z')$; as $\wp$ is even of degree 2, the fibre of $\wp$ over a non-branch value is $\{\pm z\}$, so $z' = \pm z$ (and if z is one of the four ramification points then $z' = z$ directly). Also $\wp'(z) = \wp'(z')$; if $z' = -z$ then, $\wp'$ being odd, $\wp'(z) = -\wp'(z)$, so $\wp'(z) = 0$ and z is a half-period. But the non-zero half-periods lie on ∂P for this centred parallelogram, and $z = 0$ gives $z' = -z = z$. Either way $z = z'$. Hence Φ is injective on P° and so on $\mathbb{C}/\Lambda$. □

So the two families of genus-1 surfaces we have met – complex tori and compactified plane cubics – coincide. The map Φ is what makes the addition law on a cubic curve comprehensible: it is transported from the group $\mathbb{C}/\Lambda$.

§11.7 The Field of Elliptic Functions

Theorem 11.15

Let f be an elliptic function with period lattice containing Λ, and write $\wp = \wp_\Lambda$. Then there are rational functions Q_1, Q_2 with

$$f(z) = Q_1(\wp(z)) + Q_2(\wp(z))\,\wp'(z).$$

If f is even then $Q_2 = 0$.

Proof. Case 1: f even. We first reduce to a convenient normalisation. The function f and $\wp$ each have finitely many branch points on the compact surface $\mathbb{C}/\Lambda$, so we may pick distinct $c, d \in \mathbb{C}$ which are not among the values of f at its ramification points. Replacing f by

$$\frac{f - c}{f - d},$$

which is f composed with a Möbius map, hence still even and elliptic, and from which f can be recovered by the inverse Möbius map, we may assume all the zeros and poles of f are simple and occur away from the ramification points of $\wp$.

Since f is even, its zeros come in pairs $\pm a_i$ and its poles in pairs $\pm b_j$ (modulo Λ), with $a_i \not\equiv \pm a_j$ for $i \neq j$ and similarly for the b_j; say the zeros are $\{\pm a_1, \ldots, \pm a_m\}$ and the poles $\{\pm b_1, \ldots, \pm b_n\}$. Consider

$$g(z) = \frac{(\wp(z) - \wp(a_1)) \cdots (\wp(z) - \wp(a_m))}{(\wp(z) - \wp(b_1)) \cdots (\wp(z) - \wp(b_n))}.$$

Each factor $\wp(z) - \wp(a_i)$ is elliptic of degree 2 with zeros exactly $\pm a_i$ (simple, as a_i is not a ramification point of $\wp$) and a double pole at 0. So g has exactly the same zeros and poles as f away from 0, and at 0 the pole orders are $2n - 2m$ in g and, by the valency theorem applied to f, also match. Hence f/g is an elliptic function with no zeros and no poles, so it is holomorphic, hence constant. Therefore $f = \lambda g$ is a rational function of $\wp$.

Case 2: f odd. Then $f/\wp'$ is even (as $\wp'$ is odd) and elliptic, so $f/\wp' = Q_2(\wp)$ by Case 1, giving $f = Q_2(\wp)\wp'$.

General case. Write

$$f(z) = \frac{f(z) + f(-z)}{2} + \frac{f(z) - f(-z)}{2},$$

a sum of an even and an odd elliptic function, and apply the two cases. □

In the language of fields: the field $\mathcal{M}(\mathbb{C}/\Lambda)$ of meromorphic functions on the torus is $\mathbb{C}(\wp)[\wp']$, a degree-2 extension of the rational function field $\mathbb{C}(\wp)$, with $\wp'$ satisfying the quadratic $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$. Compare Proposition 11.1, which says $\mathcal{M}(\mathbb{C}_\infty) = \mathbb{C}(z)$. In both cases the meromorphic functions on a compact Riemann surface form a field of transcendence degree 1 over $\mathbb{C}$ – this is true in general, and is the beginning of the dictionary between compact Riemann surfaces and algebraic curves.

§12 Differentials on Compact Surfaces

Corollary 5.5 told us that a compact Riemann surface carries no non-constant holomorphic function, and we responded by enlarging the target to $\mathbb{C}_\infty$. There is a second, and in the long run more important, response: keep the target $\mathbb{C}$ but change the *kind* of object we are looking at. The observation is that although $f(z)$ is not a well-defined thing to write down on a surface – it depends on the chart – the expression $f(z)\,\mathrm{d}z$ very nearly is, because the chain rule supplies exactly the factor needed to make the two chart-dependencies cancel. Objects of the form $f(z)\,\mathrm{d}z$ are called differentials, and unlike functions they exist in abundance on compact surfaces. Better still, they know the genus.

§12.1 Meromorphic Differentials

Definition 12.1 (Meromorphic Differential)

Let R be a Riemann surface. A **meromorphic differential** (or **meromorphic 1-form**) ω on R is an assignment to each chart (ϕ, U) of the conformal structure of a meromorphic function f_ϕ on $\phi(U)$, subject to the compatibility condition

$$f_\psi(w) = f_\phi(\tau(w)) \cdot \tau'(w), \qquad \tau = \phi \circ \psi^{-1},$$

for all charts $(\phi, U), (\psi, V)$ and all $w \in \psi(U \cap V)$. We write $\omega = f_\phi\,\mathrm{d}z$ for the local expression of ω in the chart ϕ, and call ω **holomorphic** if every f_ϕ is holomorphic.

The compatibility rule is nothing but the chain rule made into a definition: if $z = \tau(w)$ then $\mathrm{d}z = \tau'(w)\,\mathrm{d}w$, so

$$f_\phi(z)\,\mathrm{d}z = f_\phi(\tau(w))\,\tau'(w)\,\mathrm{d}w = f_\psi(w)\,\mathrm{d}w,$$

and the two local expressions really do describe the same object. As with charts and atlases, it suffices to prescribe the f_ϕ on the charts of some atlas: the compatibility condition then propagates the definition to the whole conformal structure.

Example 12.2 (The Sphere)

On $\mathbb{C}$, take $f_{\mathrm{id}} = 1$; the resulting differential is written $\mathrm{d}z$. Now regard it on $\mathbb{C}_\infty$. In the chart $w = 1/z$ at infinity we have $\tau(w) = 1/w$ and $\tau'(w) = -1/w^2$, so

$$\mathrm{d}z = -\frac{1}{w^2}\,\mathrm{d}w.$$

So $\mathrm{d}z$ extends to a meromorphic differential on the whole sphere, holomorphic except for a double pole at ∞. Notice that it has no zeros at all – something no meromorphic *function* on $\mathbb{C}_\infty$ can manage while having only one pole.

Example 12.3 (The Torus)

On $T = \mathbb{C}/\Lambda$ the charts of Proposition 4.26 are local inverses of π, and the transition functions are the translations $w \mapsto w + \lambda$, whose derivatives are all 1. So the constant assignment $f_\phi = 1$ is compatible, and defines a differential on T, again written $\mathrm{d}z$. It is holomorphic and has no zeros whatsoever.

Example 12.4 (The Differential of a Function)

Let f be meromorphic on R. Define $\mathrm{d}f$ by taking, in the chart (ϕ, U), the meromorphic function $(f \circ \phi^{-1})'$. Compatibility is the chain rule:

$$(f \circ \psi^{-1})'(w) = \left(f \circ \phi^{-1} \circ \tau\right)'(w) = (f \circ \phi^{-1})'(\tau(w)) \cdot \tau'(w).$$

So every meromorphic function has a differential, and $\mathrm{d}f = 0$ if and only if f is constant.

Lemma 12.5

If ω is a meromorphic differential and h a meromorphic function then $h\omega$ (defined chartwise by $hf_\phi \circ \phi^{-1}$) is a meromorphic differential. Conversely, if $\omega \not\equiv 0$ and ω' is any meromorphic differential, then $\omega' = h\omega$ for a unique meromorphic function h on R.

Proof. The first statement is immediate, since multiplying both sides of the compatibility rule by h preserves it. For the second, define h in the chart ϕ to be f'_ϕ / f_ϕ, where f_ϕ, f'_ϕ are the local expressions of ω, ω'. This is meromorphic (the denominator is not identically zero, by the identity theorem applied on the connected surface R), and the transition factors τ' cancel in the quotient, so the local definitions agree on overlaps and glue to a meromorphic function on R. Uniqueness is clear. □

So the meromorphic differentials form a one-dimensional vector space over the field of meromorphic functions – there are exactly as many differentials as there are functions, once you have found one. What makes them worth introducing is that the *zeros and poles* of a differential are counted differently from those of a function.

Definition 12.6 (Order)

Let ω be a meromorphic differential, not identically zero, and $p \in R$. The **order** of ω at p is

$$\mathrm{ord}_p(\omega) = \mathrm{ord}_{\phi(p)}(f_\phi)$$

for any chart (ϕ, U) about p, where ord denotes the order of vanishing (so a pole of order m has $\mathrm{ord} = -m$).

This is well defined: the transition factor τ' is holomorphic and nowhere zero, so multiplying by it changes no order of vanishing. We call p a zero of ω if $\mathrm{ord}_p(\omega) > 0$ and a pole if $\mathrm{ord}_p(\omega) < 0$; by the identity theorem these form a discrete set, hence a finite one when R is compact.

Definition 12.7 (Divisor and Degree)

Let R be compact. The **divisor** of a non-zero meromorphic differential ω, or of a non-constant meromorphic function f, is the formal sum

$$\mathrm{div}(\omega) = \sum_{p \in R} \mathrm{ord}_p(\omega) \cdot p, \qquad \mathrm{div}(f) = \sum_{p \in R} \mathrm{ord}_p(f) \cdot p,$$

and its **degree** is the integer $\deg \operatorname{div}(\omega) = \sum_p \operatorname{ord}_p(\omega)$.

Lemma 12.8

Let f be a non-constant meromorphic function on a compact Riemann surface R. Then $\deg \operatorname{div}(f) = 0$.

Proof. View f as a map $R \to \mathbb{C}_\infty$. By the valency theorem the sum of the orders of the zeros of f is

$$\sum_{p \in f^{-1}(0)} \operatorname{mult}_f(p) = \deg f,$$

and likewise the sum of the orders of the poles is $\sum_{p \in f^{-1}(\infty)} \operatorname{mult}_f(p) = \deg f$. These cancel. □

In other words, a meromorphic function on a compact surface has exactly as many zeros as poles. The corresponding count for a differential is not zero, and what it is instead is the whole point.

Theorem 12.9 (Degree of a Canonical Divisor)

Let R be a compact Riemann surface of genus g carrying a non-constant meromorphic function. Then for every non-zero meromorphic differential ω on R,

$$\deg \operatorname{div}(\omega) = 2g - 2.$$

Proof. *The answer does not depend on ω.* If ω' is another non-zero differential then $\omega' = h\omega$ for a meromorphic h by Lemma 12.5, and orders add: $\operatorname{ord}_p(\omega') = \operatorname{ord}_p(h) + \operatorname{ord}_p(\omega)$. Summing and applying Lemma 12.8 gives $\deg \operatorname{div}(\omega') = \deg \operatorname{div}(\omega)$. (If h is constant this is trivial.)

A convenient choice. Let $f : R \to \mathbb{C}_\infty$ be a non-constant meromorphic function, of degree n say, and take $\omega = \mathrm{d}f$, which is not identically zero. We compute $\operatorname{ord}_p(\mathrm{d}f)$ using the local normal form (Proposition 9.1), writing $m = \operatorname{mult}_f(p)$.

If $f(p) \neq \infty$, choose the chart $\psi(w) = w - f(p)$ downstairs and a chart ϕ upstairs putting f in normal form, so that $f \circ \phi^{-1}(z) = f(p) + z^m$. Then $\mathrm{d}f$ has local expression mz^{m-1}, and

$$\operatorname{ord}_p(\mathrm{d}f) = m - 1.$$

If $f(p) = \infty$, use the chart $\psi(w) = 1/w$ downstairs, so that normal form reads $1/f \circ \phi^{-1}(z) = z^m$, i.e. $f \circ \phi^{-1}(z) = z^{-m}$. Then $\mathrm{d}f$ has local expression $-mz^{-m-1}$, and

$$\operatorname{ord}_p(\mathrm{d}f) = -m - 1.$$

Comparing the two, a pole of f contributes $-m - 1 = (m - 1) - 2m$. Hence

$$\deg \operatorname{div}(\mathrm{d}f) = \sum_{p \in R} (\operatorname{mult}_f(p) - 1) - 2 \sum_{p \in f^{-1}(\infty)} \operatorname{mult}_f(p) = \sum_{p \in R} (\operatorname{mult}_f(p) - 1) - 2n,$$

the last step by the valency theorem. Finally Riemann–Hurwitz (Theorem 10.7) applied to $f : R \to \mathbb{C}_\infty$, whose target has genus 0, gives

$$2g - 2 = n(0-2) + \sum_{p \in R} (\mathrm{mult}_f(p) - 1),$$

so that $\sum_p(\mathrm{mult}_f(p) - 1) = 2g - 2 + 2n$. Substituting, $\deg \mathrm{div}(\mathrm{d}f) = 2g - 2$. □

Remark. The hypothesis that R carries a non-constant meromorphic function looks like a blemish, and it is: it is a (hard) theorem that every compact Riemann surface does, so the hypothesis is vacuous. All the surfaces we have met come with one for free, and we shall not use the general result.

Divisors of the form $\mathrm{div}(\omega)$ are called **canonical divisors**, and the theorem says they all have the same degree, namely $2g - 2$. This is a genuinely new way to compute the genus: exhibit a single differential and count its zeros and poles.

Example 12.10

On $\mathbb{C}_\infty$, the differential $\mathrm{d}z$ of Example 12.2 has divisor $-2\cdot\infty$, of degree $-2 = 2\cdot 0 - 2$. On $\mathbb{C}/\Lambda$, the differential $\mathrm{d}z$ of Example 12.3 has empty divisor, of degree $0 = 2\cdot 1 - 2$. Both agree with Theorem 12.9, as they had better.

Example 12.11 (The Cubic Curve Again)

Let $\hat{R}$ be the compactified curve $w^2 = z^3 - z$ of Section 8.3, and consider

$$\omega = \frac{\mathrm{d}z}{w}.$$

We claim ω is holomorphic and nowhere vanishing, so that $\deg \mathrm{div}(\omega) = 0$ and hence $g(\hat{R}) = 1$ – recovering Example 10.10 without any Riemann–Hurwitz bookkeeping.

Differentiating the equation gives $2w\,\mathrm{d}w = (3z^2 - 1)\mathrm{d}z$, so that

$$\omega = \frac{\mathrm{d}z}{w} = \frac{2\,\mathrm{d}w}{3z^2 - 1}$$

wherever both expressions make sense. Away from the points with $w = 0$ or $z = \infty$, the coordinate z is a chart and $w \neq 0$, so the first expression is holomorphic and non-vanishing. At each of the three points $(z_0, 0)$ with $z_0 \in \{0, \pm 1\}$ the coordinate w is a chart, and $3z_0^2 - 1 \neq 0$ there (indeed $3z^2 - 1$ and $z^3 - z$ have no common root), so the second expression is holomorphic and non-vanishing. Finally at ∞ we use the coordinates $u = 1/z$, $v = z/w$ of Section 8.3, in which the curve reads $u = v^2(1 - u^2)$; the implicit function theorem makes v a local coordinate with $u = v^2 + O(v^6)$, so $\mathrm{d}u = (2v + O(v^5))\mathrm{d}v$ and

$$\omega = \frac{\mathrm{d}z}{w} = \frac{-\mathrm{d}u/u^2}{1/(uv)} = -\frac{v\,\mathrm{d}u}{u} = -\frac{v(2v + O(v^5))}{v^2 + O(v^6)}\mathrm{d}v = \left(-2 + O(v^4)\right)\mathrm{d}v,$$

again holomorphic and non-vanishing. So $\mathrm{div}(\omega) = 0$ and $g(\hat{R}) = 1$.

§12.2 The Residue Theorem

Differentials, not functions, are the objects one integrates. If γ is a path lying inside a single chart (ϕ, U), we may define

$$\int_\gamma \omega = \int_{\phi\circ\gamma} f_\phi(z)\,\mathrm{d}z,$$

and the change of variables formula for contour integrals shows this does not depend on the chart: substituting $z = \tau(w)$ turns $f_\phi(z)\,\mathrm{d}z$ into $f_\phi(\tau(w))\tau'(w)\,\mathrm{d}w = f_\psi(w)\,\mathrm{d}w$, which is exactly the compatibility rule. A general path is cut into finitely many such pieces. So $\int_\gamma \omega$ makes sense for any path, and this is what a differential is *for*.

Definition 12.12 (Residue)

Let ω be a meromorphic differential on R and $p \in R$. The **residue** of ω at p is

$$\mathrm{res}_p(\omega) = \frac{1}{2\pi i}\int_\gamma \omega,$$

where γ is a small positively oriented loop about p inside a chart, enclosing no other pole of ω.

Equivalently, $\mathrm{res}_p(\omega)$ is the coefficient of $(z-\phi(p))^{-1}$ in the Laurent expansion of f_ϕ; the point of the integral formulation is that it makes chart-independence obvious, whereas the coefficient of z^{-1} looks as though it ought to depend on the coordinate. (It is worth pausing on this. For a meromorphic *function* the coefficient of z^{-1} is *not* chart-independent, and 'the residue of a function at a point of a Riemann surface' is meaningless. Only differentials have residues.)

Theorem 12.13 (Residue Theorem on a Compact Surface)

Let ω be a meromorphic differential on a compact Riemann surface R. Then

$$\sum_{p\in R} \mathrm{res}_p(\omega) = 0,$$

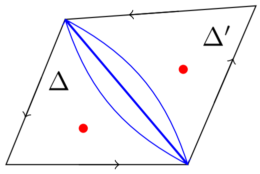
the sum being finite.

Proof. The poles $p_1, \dots, p_k$ of ω are finite in number. Choose a triangulation of R (Lemma 10.3) and subdivide until every closed triangle lies inside a single chart; perturbing slightly, we may also assume no p_i lies on an edge or vertex, so each pole is interior to exactly one triangle. Orient every triangle positively, which is possible because R is orientable and the charts are holomorphic.

For a triangle Δ contained in the chart (ϕ, U), the classical residue theorem applied to f_ϕ on the region $\phi(\Delta)$ gives

$$\int_{\partial\Delta} \omega = 2\pi i \sum_{p_i\in\Delta^\circ} \mathrm{res}_{p_i}(\omega).$$

Summing over all triangles, the right-hand side is $2\pi i \sum_p \mathrm{res}_p(\omega)$, since each pole lies in exactly one triangle. On the left, each edge of the triangulation belongs to exactly two triangles and is traversed once in each direction, so the contributions cancel in pairs and the total is 0. ☐



The picture is the whole proof: whatever ω does inside the triangles, the boundary integrals cancel along every shared edge, so their total vanishes – and their total is $2\pi i$ times the sum of all the residues.

Example 12.14

Take $R = \mathbb{C}_\infty$ and $\omega = f(z)\,\mathrm{d}z$ for a rational f. The theorem says the sum of the residues of f over $\mathbb{C}$, together with the residue at ∞ (which in the chart $w = 1/z$ is the residue at 0 of $-f(1/w)/w^2$), is zero. This is the familiar 'sum of residues including infinity vanishes' of IB Complex Analysis, now seen as a statement about the sphere.

Example 12.15

Take $R = \mathbb{C}/\Lambda$ and $\omega = f\,\mathrm{d}z$ for an elliptic function f, using the nowhere-vanishing $\mathrm{d}z$ of Example 12.3; then $\mathrm{res}_p(\omega)$ is the ordinary residue of f. The theorem says the residues of an elliptic function in a period parallelogram sum to zero, which is exactly what the cancellation of opposite sides gave us in the second proof of Corollary 11.7. The triangulation argument above is the same cancellation, done invariantly.

§12.3 The Degree–Genus Formula

We met the degree–genus formula in Example 10.12 as an observation about Fermat curves. It holds for every smooth plane curve, and the proof is a pretty application of Riemann–Hurwitz.

Theorem 12.16 (Degree–Genus Formula)

Let $C \subseteq \mathbb{P}^2$ be a smooth projective plane curve of degree $d \geq 1$. Then C is a compact Riemann surface of genus

$$g(C) = \frac{(d-1)(d-2)}{2}.$$

Proof sketch. That C is a compact Riemann surface is Proposition 9.17 together with the gluing of charts at infinity, exactly as in Section 8.3.

Pick a point $q \in \mathbb{P}^2 \setminus C$ and project from it: each $p \in C$ lies on a unique line through q, and the set of such lines is a copy of $\mathbb{P}^1 \cong \mathbb{C}_\infty$. This gives a holomorphic map

$$\pi_q : C \longrightarrow \mathbb{C}_\infty.$$

A line meets C in d points counted with multiplicity (Bézout's theorem), so $\deg \pi_q =$

d. A point p is a ramification point of π_q precisely when the line qp meets C at p with multiplicity at least 2, that is, when the line is tangent to C at p.

For a generic choice of q – and this is the step we are not proving – every tangent line through q is tangent at exactly one point and to order exactly 2, so each ramification point has multiplicity 2; and the number of such points is $d(d-1)$. (This last count is again Bézout: the points of C whose tangent passes through q are the intersections of C with its *polar curve* with respect to q, which has degree $d-1$.) Riemann–Hurwitz now gives

$$2g(C) - 2 = d\,(0-2) + d(d-1)\cdot(2-1) = d^2 - 3d,$$

whence $g(C) = (d^2 - 3d + 2)/2 = (d-1)(d-2)/2$. □

The two computations we did by hand are the special cases we can check independently: the Fermat curve $x^d + y^d = 1$ of Example 10.12, where we projected along a coordinate axis rather than from a general point and found the same answer, and the cubic $w^2 = z^3 - z$, which after homogenising is a smooth plane cubic of genus $(3-1)(3-2)/2 = 1$.

Remark. It is worth being clear about what the formula does and does not say. Every smooth plane curve of degree d has genus $(d-1)(d-2)/2$, so smooth plane curves realise only the genera $0, 0, 1, 3, 6, 10, \ldots$. Compact Riemann surfaces of the missing genera certainly exist – the hyperelliptic curves supply all of them – but they cannot be embedded in $\mathbb{P}^2$ without singularities. Genus 2 is the first example. Plane curves are special, and it is exactly the interplay between the abstract theory and such concrete models that makes the subject work.

§13 Quotients and Uniformisation

We finish by classifying all Riemann surfaces, at least in principle. The classification is startlingly clean: there are only three simply connected Riemann surfaces, and every other surface is a quotient of one of them.

§13.1 Quotients by Group Actions

Definition 13.1

Let a group G act on a topological space X by homeomorphisms. The action is

(i) **properly discontinuous** if for every compact $K \subseteq X$ the set $\{g \in G : g(K) \cap K \neq \emptyset\}$ is finite;

(ii) **free** if $\mathrm{Stab}_G(x) = \{1\}$ for every $x \in X$.

Example 13.2

A lattice Λ acts on $\mathbb{C}$ by translation. This action is free (a non-zero translation has no fixed point) and properly discontinuous (a compact K is bounded, and only finitely many $\lambda \in \Lambda$ have $|\lambda| \leq \operatorname{diam} K$).

Lemma 13.3

Let G act freely and properly discontinuously by homeomorphisms on a Riemann surface R. Then $G\backslash R$ is Hausdorff and the quotient map $\pi : R \to G\backslash R$ is a regular covering map.

Proof. First, π is open, since $\pi^{-1}(\pi(U)) = \bigcup_g g(U)$.

Hausdorff. Let $p, q \in R$ with $\pi(p) \neq \pi(q)$. Choose open $U \ni p$, $V \ni q$ with compact closures, and set $K = \bar{U} \cup \bar{V}$, compact. Then

$$\{g \in G : g(\bar{U}) \cap \bar{V} \neq \emptyset\} \subseteq \{g : g(K) \cap K \neq \emptyset\}$$

is finite, say $\{g_1, \dots, g_k\}$. For each i we have $g_i p \neq q$ (else $\pi(p) = \pi(q)$), so choose disjoint open $U_i \ni p$ and $V_i \ni g_i q$. Shrinking to

$$U' = U \cap \bigcap_i U_i, \qquad V' = V \cap \bigcap_i g_i^{-1}(V_i),$$

we get $g(U') \cap V' = \emptyset$ for every $g \in G$. Then $\pi(U')$ and $\pi(V')$ are disjoint open neighbourhoods.

Covering. Let $p \in R$ and choose open $U \ni p$ with $\bar{U}$ compact, so that $\{g : g(\bar{U}) \cap \bar{U} \neq \emptyset\} = \{1, g_1, \dots, g_k\}$ is finite. Freeness gives $g_i p \neq p$, so choose disjoint open $U_i \ni p$, $V_i \ni g_i p$, and set

$$U' = U \cap \bigcap_i \left(U_i \cap g_i^{-1}(V_i)\right).$$

Then $g(U') \cap U' = \emptyset$ for all $g \neq 1$, so $\pi^{-1}(\pi(U')) = \coprod_g g(U')$ and each $g(U')$ maps homeomorphically onto $\pi(U')$ (as π is open, continuous and injective on U'). So $\pi(U')$ is evenly covered. □

Proposition 13.4

Let R be a Riemann surface and G a group acting freely and properly discontinuously on R by *conformal equivalences*. Then $S = G\backslash R$ carries a unique conformal structure making $\pi : R \to S$ holomorphic, and π is then a regular holomorphic covering map.

Proof. S is connected as a continuous image of R, and Hausdorff with π a regular covering map by Lemma 13.3. Charts on S are obtained by pushing charts of R forward along local inverses of π: for U' as in the lemma and a chart (ϕ, U'') with $U'' \subseteq U'$, take $(\phi \circ (\pi|_{U'})^{-1}, \pi(U''))$. Two such charts differ by a transition function of R composed with the action of some $g \in G$, which is holomorphic by hypothesis. Uniqueness is as in Lemma 4.21. □

This is the converse of Lemma 4.21, and together they say: quotients by free properly discontinuous actions of conformal automorphisms are exactly covering spaces, in the holomorphic category. The complex torus is the case $R = \mathbb{C}$, $G = \Lambda$.

Theorem 13.5

Let R be a compact Riemann surface with $g(R) \geq 2$ and let G act freely and properly discontinuously on R by conformal equivalences. Then G is finite, with

$$|G| \leq g(R) - 1.$$

Proof. By Proposition 13.4, $S = G\backslash R$ is a Riemann surface, compact as a continuous image of R, and $\pi : R \to S$ is an unramified holomorphic map of degree $|G|$ (the fibres are the G-orbits, which all have size $|G|$ by freeness; in particular G is finite since fibres are finite). Riemann–Hurwitz with no ramification gives

$$g(R) - 1 = |G| \left(g(S) - 1 \right).$$

Since $g(R) \geq 2$ the left side is positive, so $g(S) \geq 2$ and $g(S) - 1 \geq 1$, whence $|G| \leq g(R) - 1$. □

Remark. This fails for $g = 1$: the torus $\mathbb{C}/\Lambda$ acts on itself freely by translation, and any finite subgroup does too, with no bound on its order. The genus ≥ 2 hypothesis is essential, and it is the first sign that high-genus surfaces are rigid. (The sharp statement, Hurwitz's automorphism theorem, bounds the full automorphism group by $84(g-1)$; that is beyond us here.)

§13.2 The Uniformisation Theorem

Theorem 13.6 (Uniformisation Theorem)

Every simply connected Riemann surface is conformally equivalent to exactly one of

$$\mathbb{C}_\infty, \qquad \mathbb{C}, \qquad \mathbb{D} = D(0,1).$$

The proof is well beyond this course. What we can do is check that the three are genuinely distinct, and then extract the consequences.

Proposition 13.7

No two of $\mathbb{C}_\infty, \mathbb{C}, \mathbb{D}$ are conformally equivalent.

Proof. $\mathbb{C}_\infty$ is compact and the other two are not. If $\mathbb{D} \cong \mathbb{C}$ then there would be a non-constant holomorphic map $\mathbb{C} \to \mathbb{D}$, hence a bounded non-constant entire function, contradicting Liouville. □

Corollary 13.8

Every conformal structure on S^2 is conformally equivalent to $\mathbb{C}_\infty$; equivalently, a compact Riemann surface of genus 0 is the Riemann sphere.

Proof. S^2 is simply connected, so it is one of the three; it is compact, so it is $\mathbb{C}_\infty$. □

For non-simply-connected surfaces we pass to the universal cover.

Theorem 13.9

Every Riemann surface R admits a regular covering map $\pi : \tilde{R} \to R$ with $\tilde{R}$ simply connected. Moreover $\tilde{R}$ has a unique conformal structure making π holomorphic, and the deck transformation group

$$G = \mathrm{Deck}(\pi) = \{\varphi : \tilde{R} \to \tilde{R} \text{ conformal} : \pi \circ \varphi = \pi\} \cong \pi_1(R)$$

acts freely and properly discontinuously by conformal equivalences with $G \backslash \tilde{R} \cong R$.

Proof. Existence of the universal cover and the identification $\mathrm{Deck}(\pi) \cong \pi_1(R)$ for a simply connected cover are from II Algebraic Topology (a Riemann surface is locally homeomorphic to a disc, hence locally path-connected and semi-locally simply connected, so the theory applies). The conformal structure on $\tilde{R}$ comes from Lemma 4.21, and each deck transformation is then holomorphic (locally it is $(\pi|_{\tilde{U}'})^{-1} \circ \pi$) with holomorphic inverse. Freeness and proper discontinuity are standard properties of deck groups of covering spaces. □

Corollary 13.10

Every Riemann surface R is conformally equivalent to $G \backslash \tilde{R}$, where $\tilde{R} \in \{\mathbb{C}_\infty, \mathbb{C}, \mathbb{D}\}$ and $G \cong \pi_1(R)$ acts freely and properly discontinuously by conformal automorphisms of $\tilde{R}$.

We say R is **uniformised by** $\tilde{R}$. To make this useful we need to know the conformal automorphisms of each model.

Proposition 13.11 (i) $\mathrm{Aut}(\mathbb{C}_\infty) = \{z \mapsto \frac{az+b}{cz+d} : ad - bc \neq 0\} \cong \mathrm{PGL}_2(\mathbb{C})$, the Möbius group.

(ii) $\mathrm{Aut}(\mathbb{C}) = \{z \mapsto az + b : a \in \mathbb{C}^*, b \in \mathbb{C}\}$.

(iii) $\mathrm{Aut}(\mathbb{D}) = \{z \mapsto e^{i\theta} \frac{z-a}{1-\bar{a}z} : a \in \mathbb{D}, \theta \in \mathbb{R}\}$.

Proof sketch. (i) A conformal automorphism of $\mathbb{C}_\infty$ is a degree-1 meromorphic function, hence rational of degree 1 by Proposition 11.1, hence Möbius. (ii) An automorphism of $\mathbb{C}$ extends over ∞: by Casorati–Weierstrass the singularity at ∞ cannot be essential (an injective function cannot have dense image on every punctured neighbourhood while also being injective near a distant point), so it is a pole or removable; removable would force boundedness and constancy, so it is a pole, and f is a polynomial, which is injective only if it is linear. (iii) This was proved in IB Complex Analysis using the Schwarz lemma. □

Proposition 13.12

If R is uniformised by $\mathbb{C}_\infty$ then $R \cong \mathbb{C}_\infty$.

Proof. Say $R = G \backslash \mathbb{C}_\infty$ with G acting freely by Möbius maps. But every Möbius map has a fixed point in $\mathbb{C}_\infty$: the equation $az + b = z(cz + d)$ is a quadratic (or linear) with a root in $\mathbb{C}$, unless $c = 0$ and $a = d$, in which case $z \mapsto z + b/d$ fixes ∞.

So freeness forces $G = \{1\}$. □

Proposition 13.13

If R is uniformised by $\mathbb{C}$, then exactly one of the following holds:

(i) $G = \{1\}$ and $R \cong \mathbb{C}$;

(ii) $G \cong \mathbb{Z}$ and $R \cong \mathbb{C}^*$;

(iii) $G \cong \mathbb{Z}^2$ and $R \cong \mathbb{C}/\Lambda$ for a lattice Λ.

Proof. By Proposition 13.11(ii), G consists of maps $z \mapsto az + b$. If $a \neq 1$ then $z = b/(1-a)$ is a fixed point, so freeness forces $a = 1$: every element of G is a translation. Identifying G with a subgroup of $(\mathbb{C}, +)$, proper discontinuity forces G to be discrete, and a discrete subgroup of $\mathbb{C}$ is trivial, infinite cyclic, or a lattice (Lemma 11.3). These give $\mathbb{C}$, $\mathbb{C}/\langle\omega\rangle \cong \mathbb{C}^*$ (via $z \mapsto e^{2\pi i z/\omega}$), and $\mathbb{C}/\Lambda$. □

Lemma 13.14 (Lifting Lemma)

Let $f : R \to S$ be holomorphic with R simply connected, and let $\pi : \tilde{S} \to S$ be the universal cover of S. Then there is a holomorphic $F : R \to \tilde{S}$ with $f = \pi \circ F$.

Proof. The topological lifting criterion from II Algebraic Topology gives a continuous lift F, since $f_*\pi_1(R) = \{1\} \subseteq \pi_*\pi_1(\tilde{S})$. It is holomorphic because locally $F = (\pi|_{\tilde{U}})^{-1} \circ f$, a composite of holomorphic maps. □

Proposition 13.15

A Riemann surface is uniformised by at most one of $\mathbb{C}_\infty, \mathbb{C}, \mathbb{D}$.

Proof. The universal cover is unique up to homeomorphism, so this is really the statement that the three models are not conformally equivalent, which we know. Explicitly: suppose R were uniformised both by $\mathbb{D}$ and by $\tilde{R} \in \{\mathbb{C}, \mathbb{C}_\infty\}$, with uniformising maps $\pi : \mathbb{D} \to R$ and $f : \tilde{R} \to R$. Since $\tilde{R}$ is simply connected, Lemma 13.14 gives holomorphic $F : \tilde{R} \to \mathbb{D}$ with $f = \pi \circ F$. Restricting to $\mathbb{C} \subseteq \tilde{R}$, F is a bounded entire function, hence constant by Liouville; so f is constant, contradicting surjectivity. □

So every Riemann surface other than $\mathbb{C}_\infty, \mathbb{C}, \mathbb{C}^*$ and the tori is uniformised by the disc. In particular:

Corollary 13.16

Every compact Riemann surface of genus ≥ 2 is uniformised by $\mathbb{D}$.

Proof. It is not $\mathbb{C}_\infty$ (genus 0), and by Proposition 13.13 the only compact surface uniformised by $\mathbb{C}$ is a torus (genus 1). □

§13.3 Fuchsian Groups and the Hyperbolic Plane

The disc case is the generic one, and it connects this course to IB Geometry. Recall the Möbius map

$$z \mapsto \frac{1+iz}{1-iz}$$

carries the upper half-plane $\mathbb{H}$ conformally onto $\mathbb{D}$; under it,

$$\operatorname{Aut}(\mathbb{H}) = \left\{ z \mapsto \frac{az+b}{cz+d} : a, b, c, d \in \mathbb{R},\ ad - bc = 1 \right\} \cong \mathrm{PSL}_2(\mathbb{R}).$$

These are precisely the orientation-preserving isometries of the hyperbolic metric

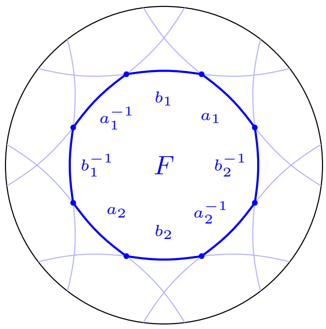
$$\frac{|\mathrm{d}z|^2}{(\operatorname{Im} z)^2}$$

on $\mathbb{H}$ (equivalently $4|\mathrm{d}z|^2/(1-|z|^2)^2$ on $\mathbb{D}$), as was shown in IB Geometry. So a Riemann surface uniformised by $\mathbb{D}$ inherits a hyperbolic metric, and its conformal geometry is hyperbolic geometry.

Definition 13.17 (Fuchsian Group)

A subgroup $\Gamma \leq \mathrm{PSL}_2(\mathbb{R})$ acting properly discontinuously on $\mathbb{H}$ is called a **Fuchsian group**.

By Corollary 13.10, the Riemann surfaces uniformised by the disc are exactly the quotients $\Gamma \backslash \mathbb{H}$ for torsion-free Fuchsian Γ. Such a quotient is understood through a **fundamental domain**: a region $F \subseteq \mathbb{D}$ whose Γ-translates tile the disc and cover the quotient exactly once. For a compact genus-2 surface one may take a regular hyperbolic octagon with all angles $2\pi/8$, whose sides are glued in the pattern $a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1}$ – the same octagon as in Section 10, but now with a genuine hyperbolic structure.



a fundamental domain in $\mathbb{D}$

§13.4 Consequences of Uniformisation

We close with two classical theorems which fall out of the machine.

Theorem 13.18 (Riemann Mapping Theorem)

Every simply connected domain $D \subsetneq \mathbb{C}$ is conformally equivalent to $\mathbb{D}$.

Proof. D is a simply connected Riemann surface, so by Theorem 13.6 it is $\mathbb{C}_\infty$, $\mathbb{C}$ or $\mathbb{D}$. It is not $\mathbb{C}_\infty$, being non-compact. Suppose $f : \mathbb{C} \to D$ were a conformal equivalence. Composing with the inclusion $D \hookrightarrow \mathbb{C}$ gives an injective entire function f; by Casorati–Weierstrass its singularity at ∞ is not essential (an injective function cannot take values densely near ∞ while being open near a fixed finite point), so f extends to a holomorphic $\bar{f} : \mathbb{C}_\infty \to \mathbb{C}_\infty$. This is non-constant, so it is surjective by Corollary 5.4; injectivity of f then forces $\bar{f}(\infty) = \infty$ and $D = f(\mathbb{C}) = \mathbb{C}$, contradicting $D \subsetneq \mathbb{C}$. So $D \cong \mathbb{D}$. □

It is worth noticing how the logic runs: uniformisation is the general theorem and the Riemann mapping theorem is a special case, whereas historically the implication is usually run the other way (the Riemann mapping theorem being an ingredient in proofs of uniformisation).

Theorem 13.19 (Little Picard Theorem)

Every holomorphic function $f : \mathbb{C} \to \mathbb{C} \setminus \{0, 1\}$ is constant.

Proof. The surface $\mathbb{C} \setminus \{0, 1\}$ is uniformised by $\mathbb{D}$: it is not $\mathbb{C}_\infty$ or $\mathbb{C}$ or $\mathbb{C}^*$ or a torus, so by Propositions 13.12 and 13.13 its universal cover is $\mathbb{D}$. (One can also exhibit the covering explicitly using the modular λ-function; this is on the example sheets.)

Let $\pi : \mathbb{D} \to \mathbb{C} \setminus \{0, 1\}$ be the uniformising map. As $\mathbb{C}$ is simply connected, Lemma 13.14 gives a holomorphic $F : \mathbb{C} \to \mathbb{D}$ with $f = \pi \circ F$. But F is a bounded entire function, hence constant by Liouville, so f is constant. □

Of course $\{0, 1\}$ may be replaced by any two distinct points of $\mathbb{C}$, by composing with a suitable affine map. The theorem is a dramatic strengthening of Liouville: an entire function which omits merely *two* values is constant. That such a statement should follow from a covering-space argument – with no estimate in sight – is a fair summary of what Riemann surfaces are for.